

DNA, Cell Division, & Inheritance Unit

- *Remember from chapter 1:*
 - life reproduces itself (with DNA), grows, & develops
- *Remember from chapter 3:*
 - all cells contain DNA that is duplicated & passed on
- Chapter 7 = How do we get from DNA to walking & talking us?
- Chapter 8 = How does DNA duplicate itself & how do we grow?
- Chapter 9 = How do we pass on DNA to reproduce?

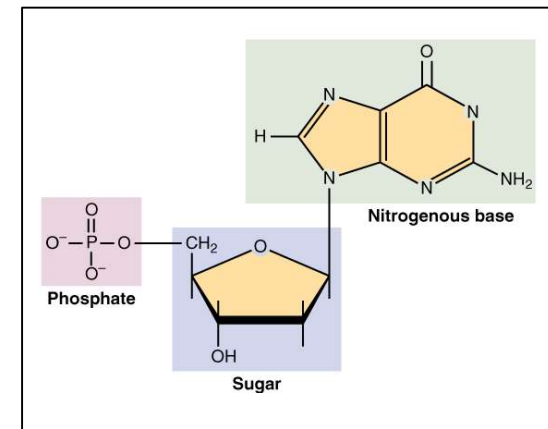
Chapter 6. Part1: DNA Structure & Gene Function

- What is the structure of DNA?
- How does DNA code for traits?
- How are genes regulated?
- How do mutations affect genes?

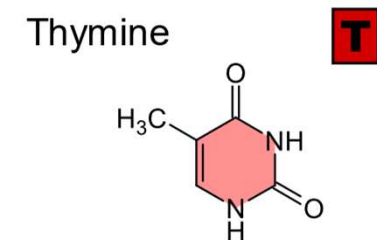
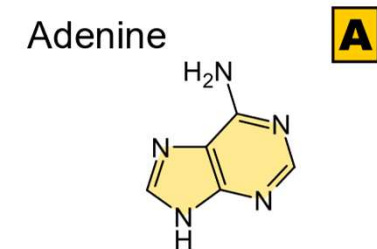
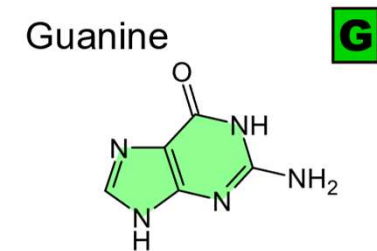
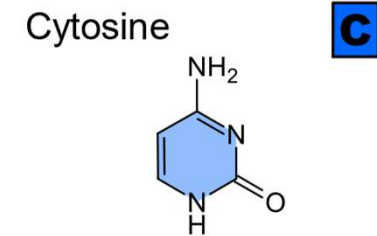
Corresponds with OpenStax Biology 2e Chapters 14,15 & 16

What is the structure of DNA?

- *Remember:* **DNA** (deoxyribonucleic acid) is a nucleic acid (polymer) made up of nucleotides (monomers)
- *Remember:* all nucleotides have at least 1 phosphate, a sugar, & a base
 - In DNA: always 1 phosphate & always same sugar (deoxyribose)
 - Base can be any 1 of 4 though

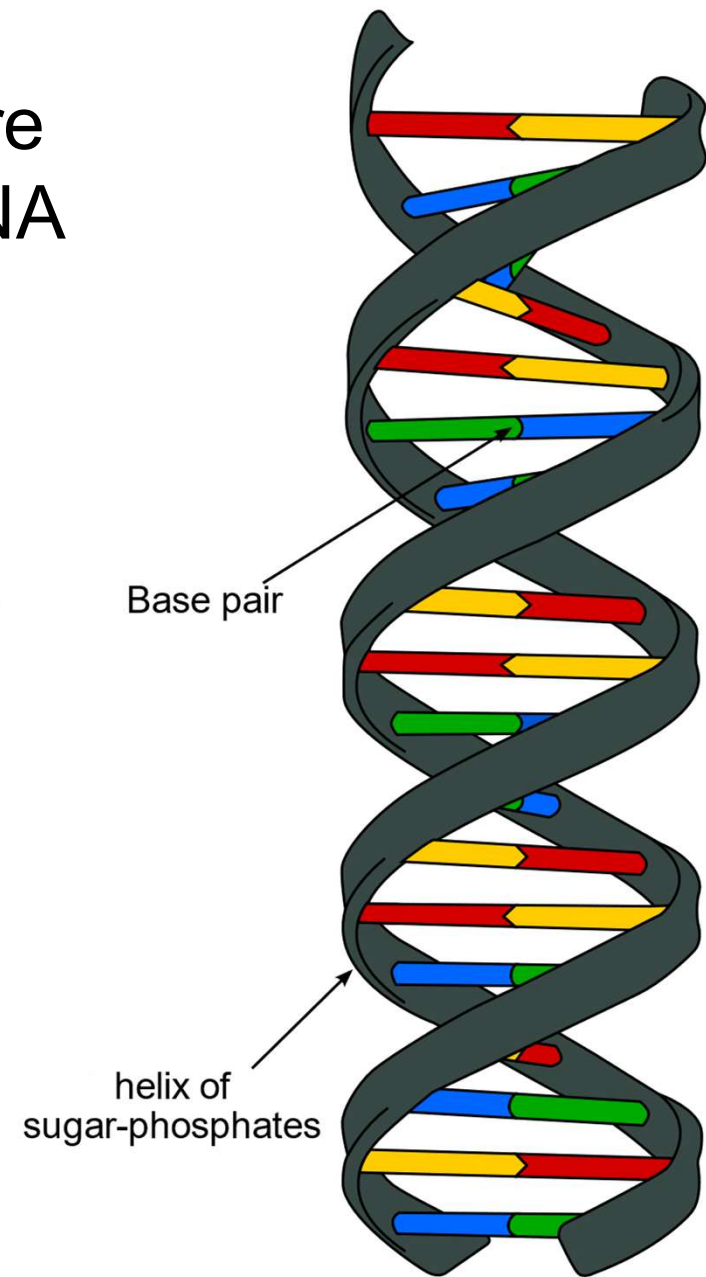


- A nucleotide can have 1 of 4 bases
 - Adenine: A
 - Guanine: G
 - Thymine: T
 - Cytosine: C
- Even though it's only the base that changes, the entire nucleotide is often referred to by the letter representing the base (A, G, T, or C)



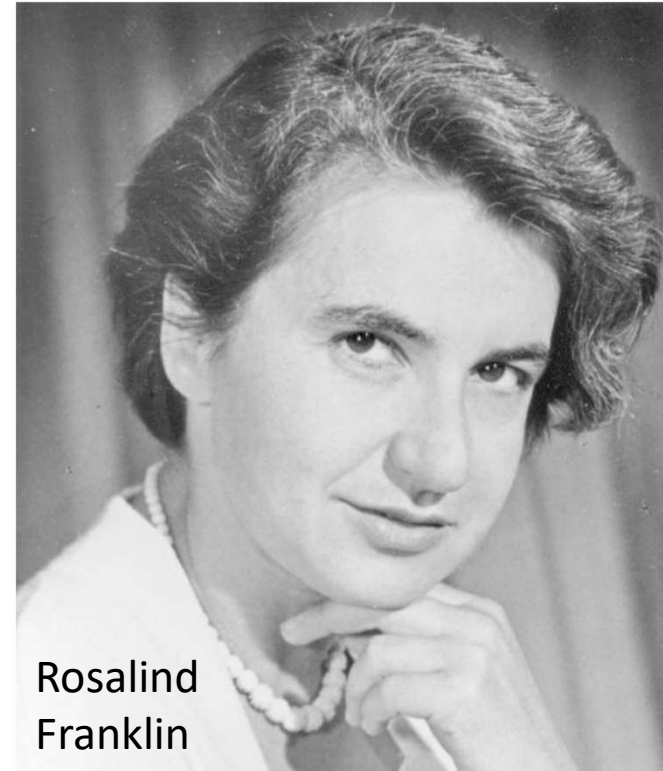
Nucleobases
of DNA

- The secrets of DNA function are found in the 3D structure of DNA
 - DNA = a **double helix**
 - **Double** = two strands connect together (like a ladder)
 - **Helix** = the nucleotide strands are then twisted like a corkscrew or circular staircase





(a)



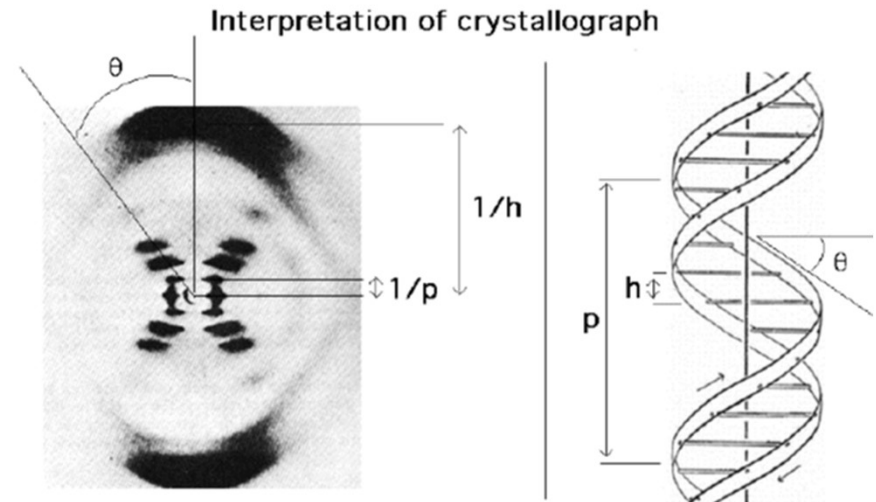
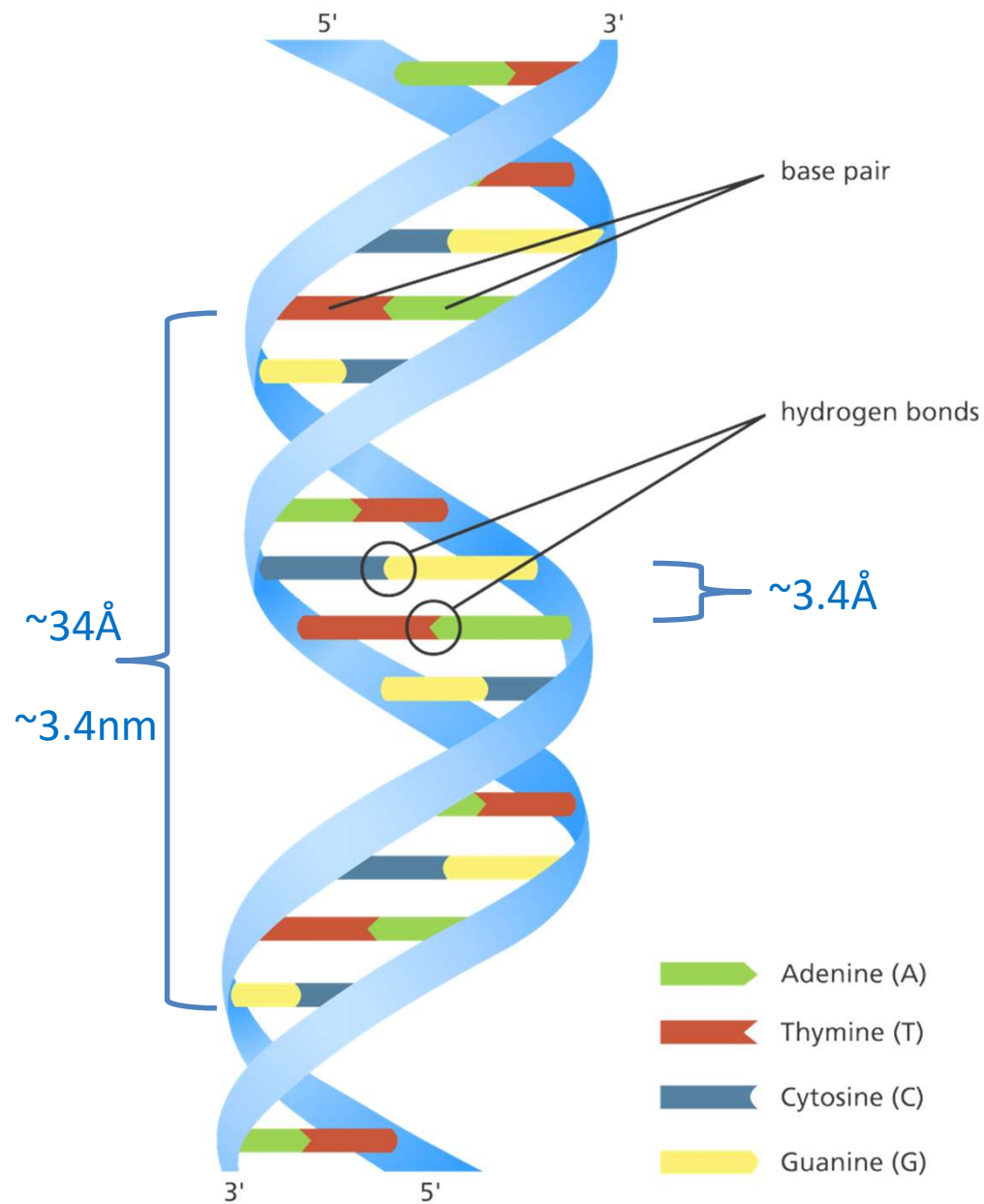
Rosalind
Franklin

(b)

The work of pioneering scientists (a) James Watson, Francis Crick, and Maurice Wilkins led to our understanding of DNA. (b) the X-ray diffraction pattern of DNA obtained by Rosalind Franklin in 1950s, which helped to elucidate its double-helix structure

James Watson et al., awarded Nobel Prize in Physiology or Medicine in 1962

He died on Nov 6, 2025 at the age of 97 (Franklin died in 1958, a few years after discovery)

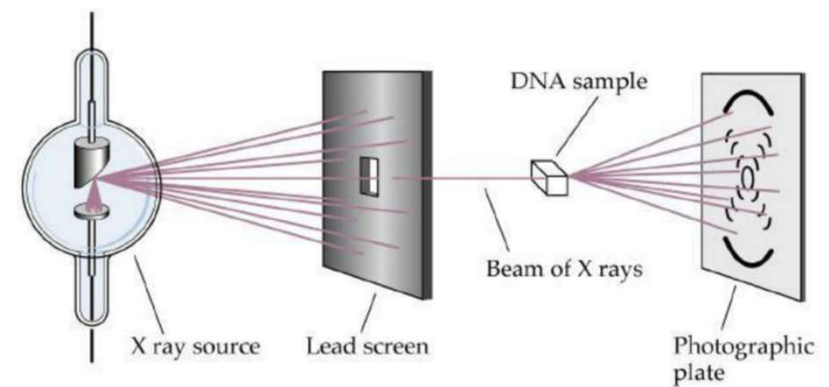


θ - tilt of helix (angle from perpendicular to long axis)

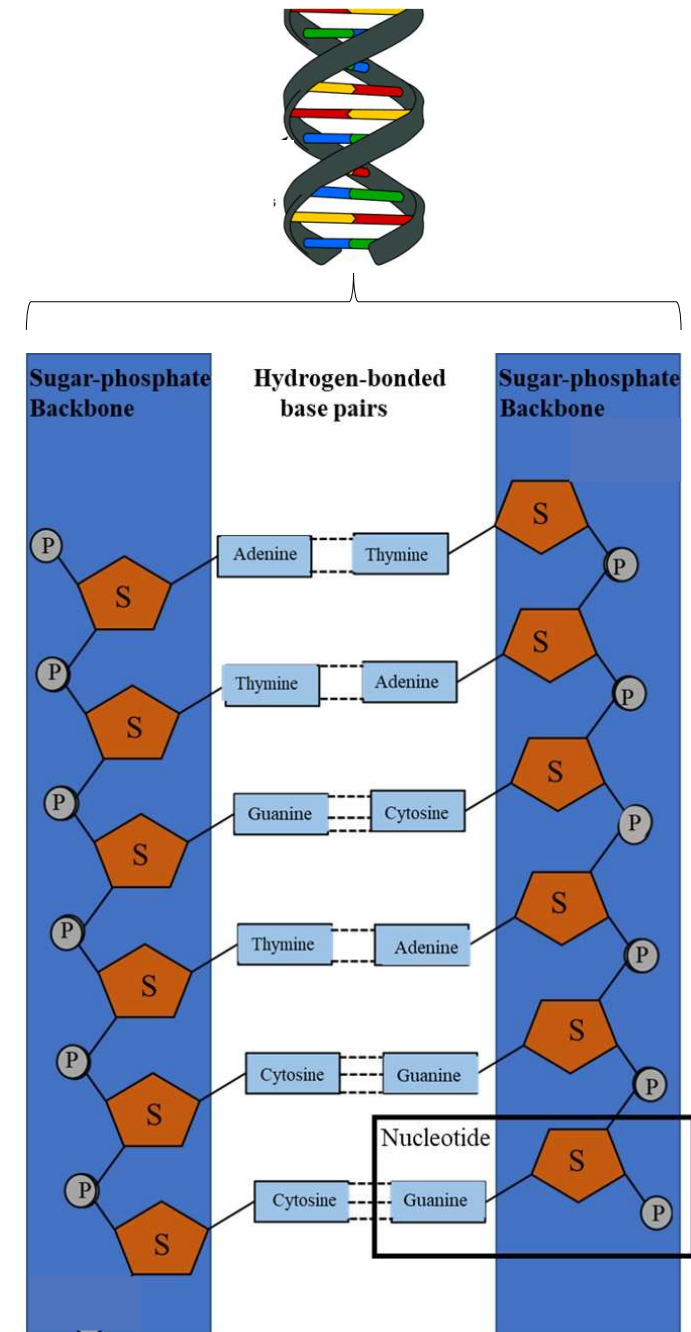
$h = 3.4 \text{ \AA}$ (Distance between bases)

$p = 34 \text{ \AA}$ (Distance for one complete turn of helix;
Repeat unit of the helix)

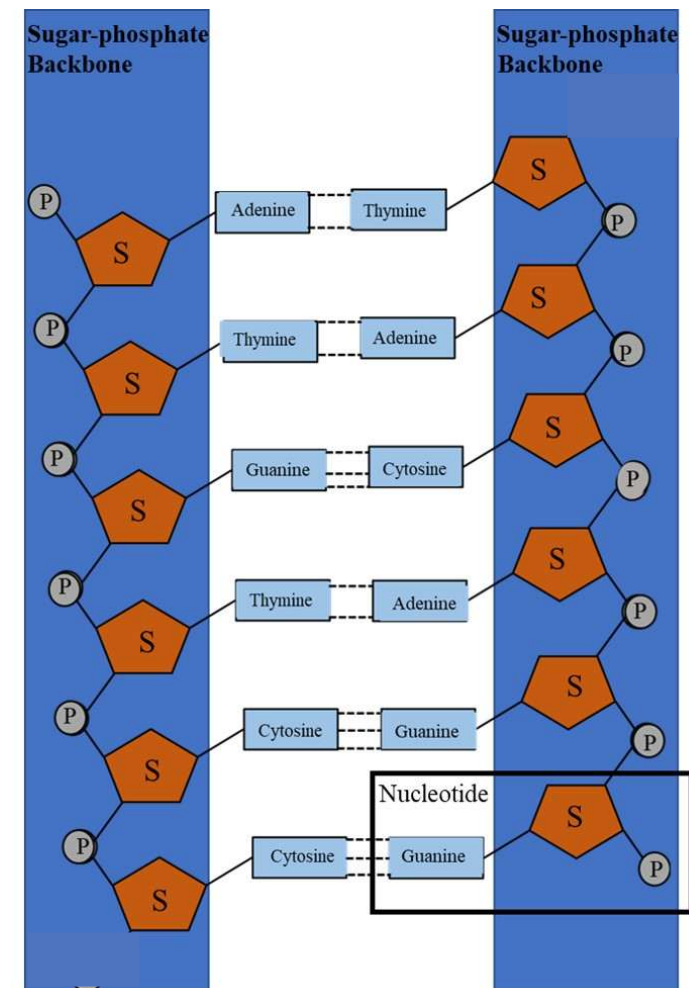
X-Ray crystallography



- Within each DNA strand, the phosphate group of one nucleotide binds to the sugar of the next nucleotide in the same strand
 - Bond is covalent (strong!)
 - The sugar & phosphate portions of DNA make up the **sugar-phosphate backbone**
 - The 2 strands run **antiparallel** (in opposite directions)



- The 2 strands are held together through bonds between the bases (making the “rungs” of the ladder)
 - Hydrogen bonds hold bases together (weak)
 - Bases pair in an exact combination: **complimentary base pairs**
 - A always binds with T
 - G always binds with C



Length and Width of a Human DNA Molecule

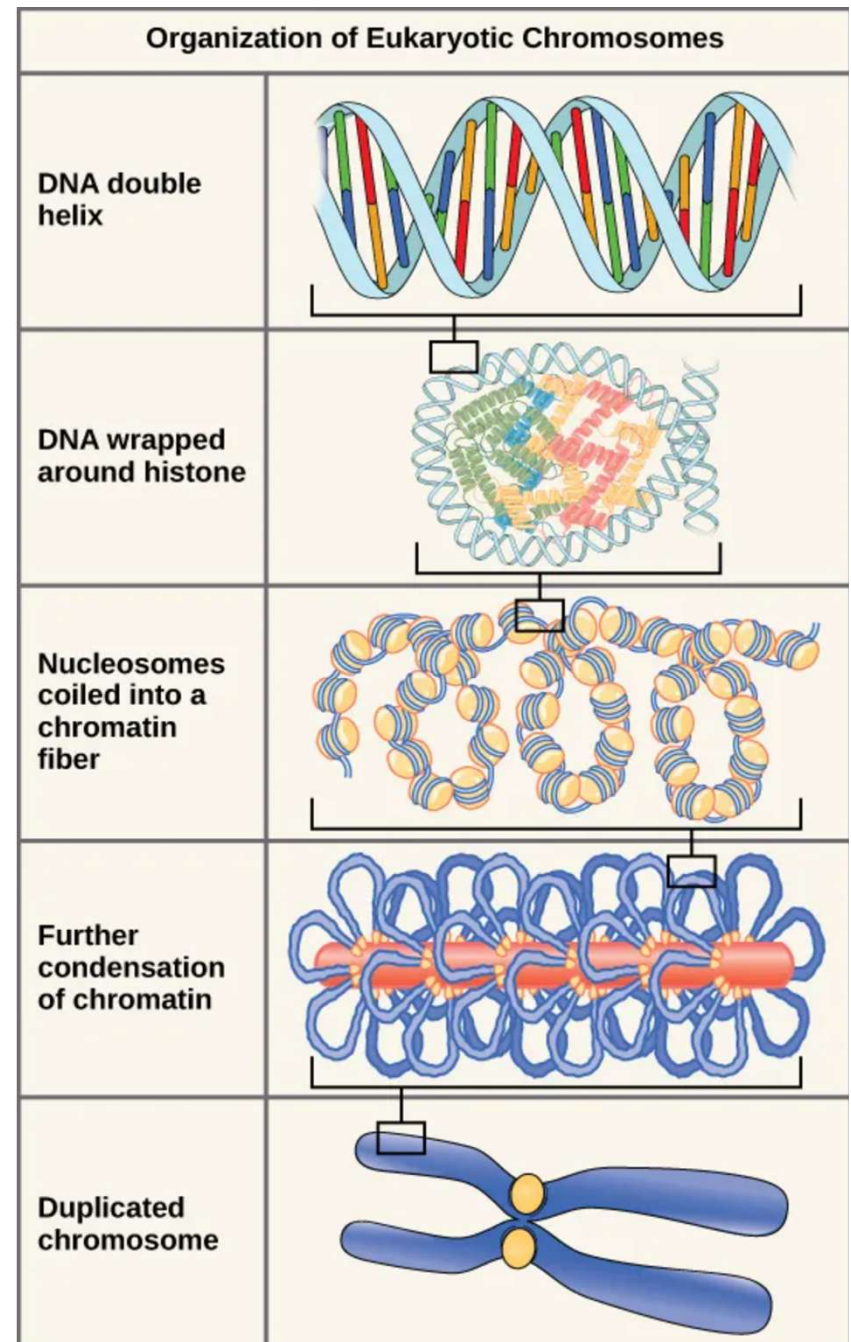
Bibliographic Entry	Result (w/surrounding text)	Standardized Result
Mitchel, Campbell Reece. <i>Biology Concept and Connections</i> . California, 1997.	"At actual size, a human cell's DNA totals about 3 meters in length."	3.0 m
McGraw Hill <i>Encyclopedia of Science and Technology</i> . New York: McGraw Hill, 1997.	"If stretched out, would form very thin thread, about 6 feet (2 meters) long."	2.0 m
Matthews, Harry R. <i>DNA Structure Prerequisite Information</i> . 1997.	"The length is (length of 1 bp)(number of bp per cell) which is (0.34 nm)(6×10^9)"	2.0 m
Leltinger, Albert L. <i>Biochemistry</i> . New York: Worth, 1975.	"Chromosome 13 contains a DNA molecule about 3.2 cm long."	1.5 m
"Cell." <i>The World Book Encyclopedia</i> . Chicago: Field Enterprises, 1996.	"On the average, a single human chromosome consists of DNA molecule that is about 2 inches long."	2.3 m

Width = ~2.5 nanometers

For 2 meter DNA, how many base-pairs?

Genome Project

- Packing of eukaryotic chromosome



■ Human Genome Project - International Research Initiative:

The Human Genome Project (HGP) was an international research initiative launched in October 1990 and completed in April 2003. It costed ~US\$2.7 billion. Its primary goal was to sequence the 3 billion bases of the human genome, providing a comprehensive map of human genetic material.

This monumental achievement has significantly advanced our understanding of human biology, genetics, and medicine, leading to improved diagnostic and therapeutic strategies.

Additionally, the project addressed ethical, legal, and social implications of genomic research, ensuring responsible use of genetic information. Overall, the HGP is regarded as one of the greatest scientific achievements in history, reshaping our approach to health and disease.

Triggered development of technologies for gene analysis and related diagnostic techniques. (also for plants, animals, viruses)

- DNA Sequencing: Reading the sequence of DNA

General reading: Chapter 14.2

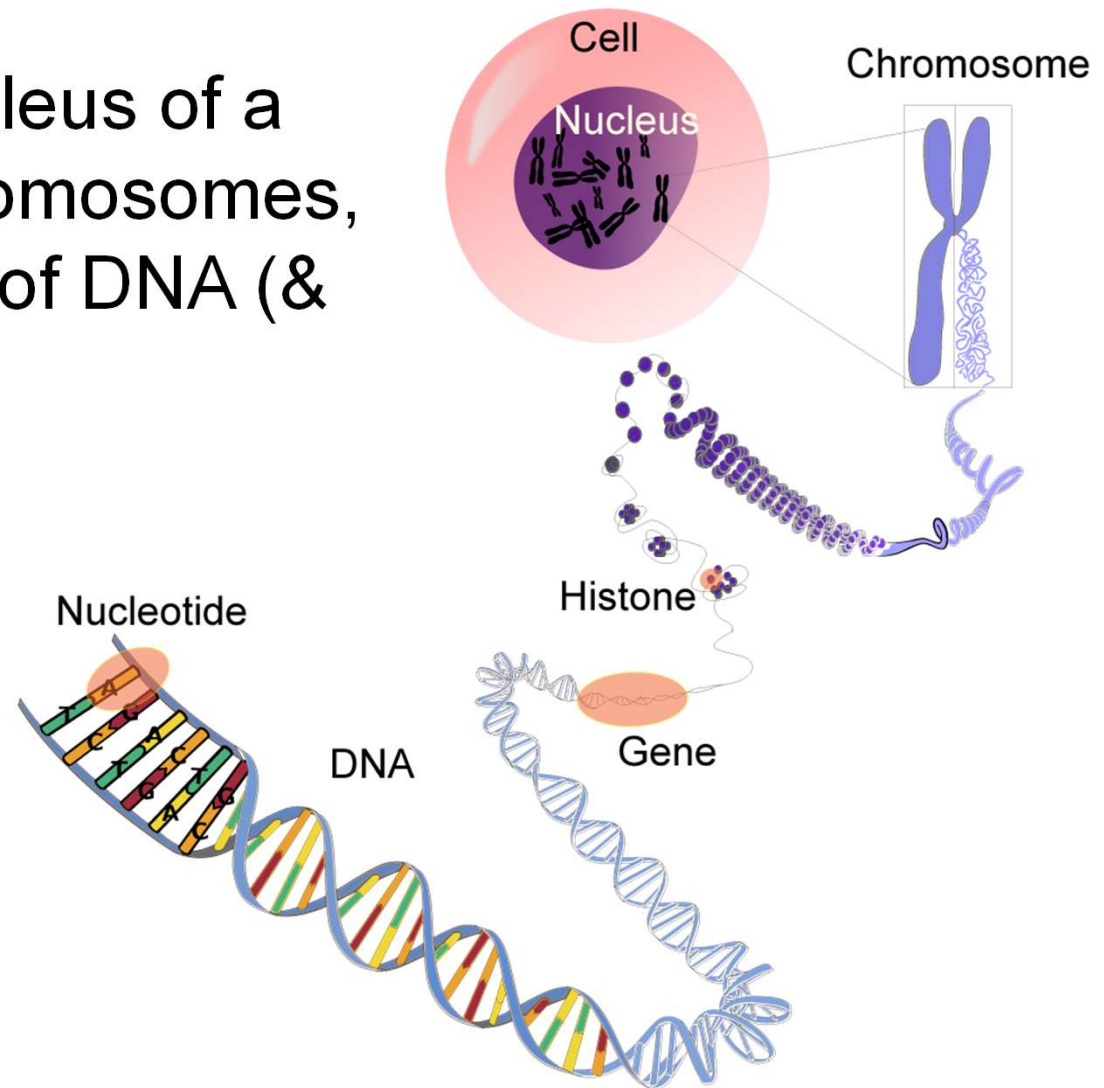
Biology Animation Library

Early DNA Sequencing:

Sanger Sequencing

How does DNA code for traits?

- *Remember:* the nucleus of a cell contains its chromosomes, which are made up of DNA (& proteins)
 - Sequences of DNA that code for a specific protein/trait = **genes**

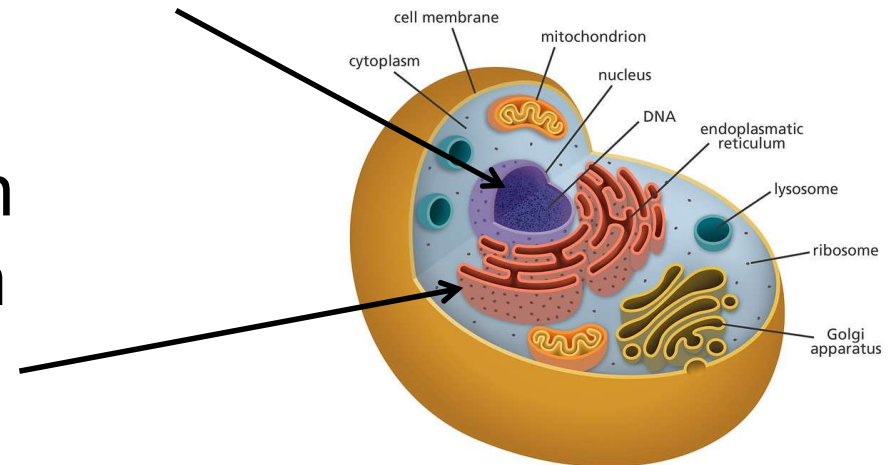


- Genes are usually hundreds or thousands of nucleotides long (ATGCCTAAATCGGTA....)
- Genes code for specific proteins, which is how we get our **traits**
 - e.g. our skin, hair, & eye color (traits) are caused by melanin pigment proteins that are coded for by genes
 - e.g. our ability to see the colors red & green (a trait) is caused by a protein in the eye, which is coded for by a gene

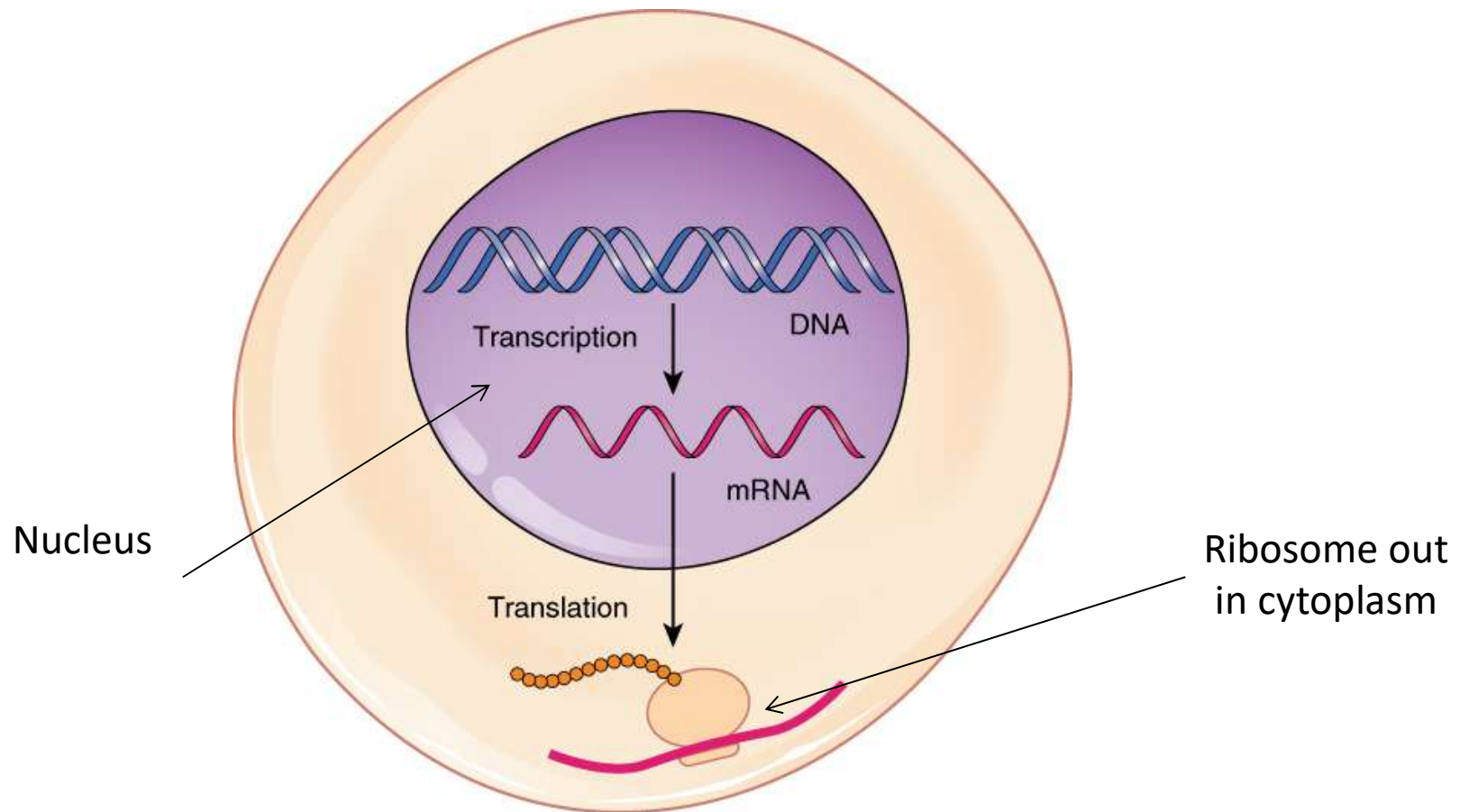


A fruit stand with normal color vision and simulated red-green colorblindness

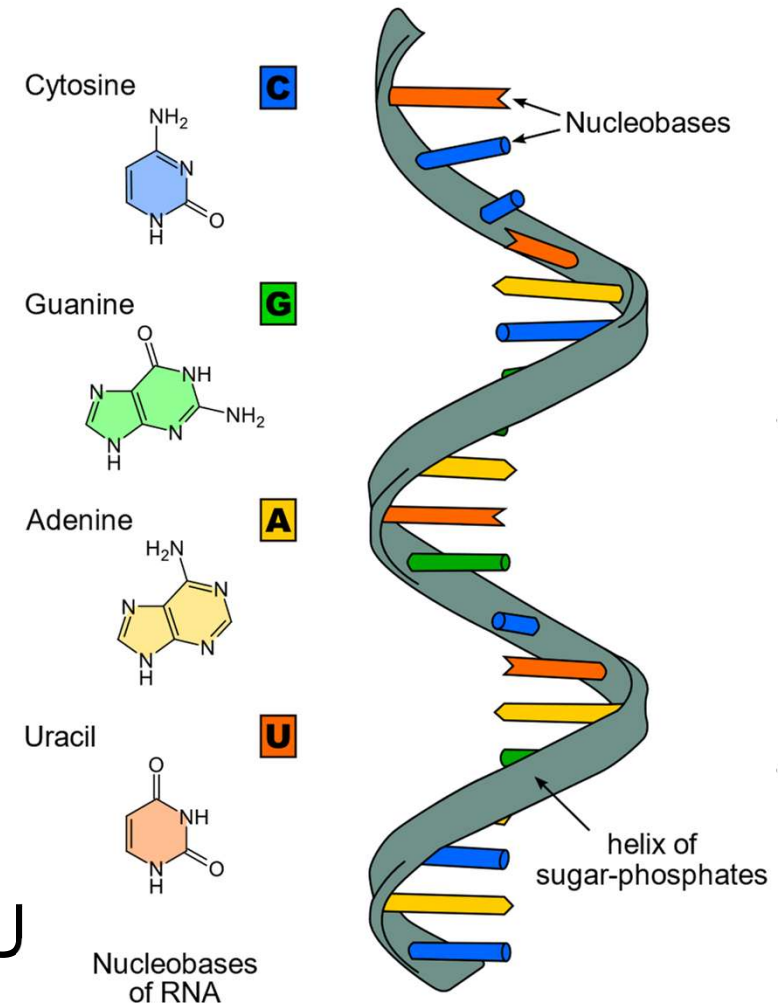
- Proteins do nearly everything, including make up us
- DNA holds the blueprints, coding the information on what kind of proteins (& how many of each) to make
 - So we know that to make up walking & talking us, there is a flow of information from DNA → proteins
 - HOWEVER, there is a problem:
 - DNA is locked inside the nucleus
 - Proteins are made at ribosomes & the rough endoplasmic reticulum **OUTSIDE** the nucleus



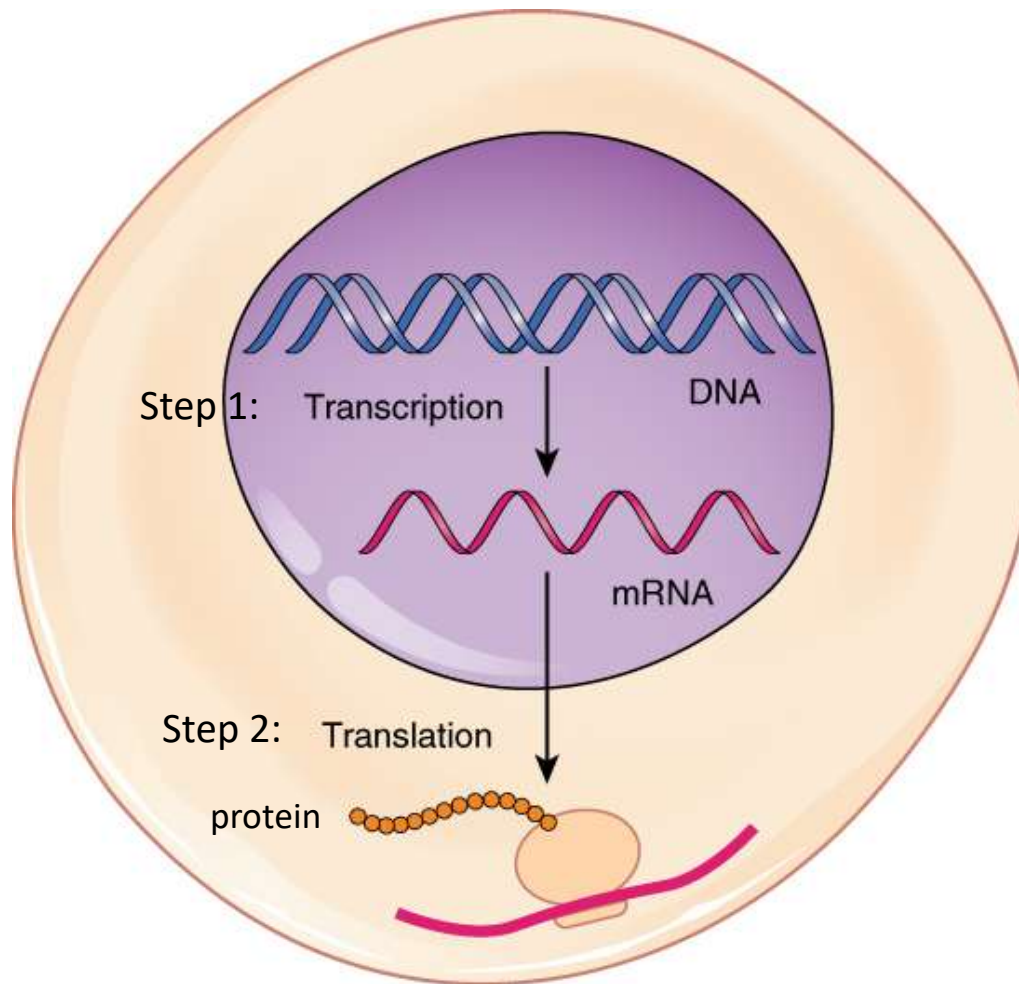
- We must have a middle man to take the information from DNA & transfer the instructions to proteins
 - the middle man = **RNA**



- **RNA** (ribonucleic acid) is a nucleic acid like DNA, but differs in a few key ways:
 - has a different sugar (ribose)
 - is single stranded
 - does not contain the base T instead has the base U (uracil)
 - so RNA bases = A U C G
 - in RNA, G still pairs with C but A pairs with U



- Getting from DNA to proteins is a 2 step process



Step 1: DNA → mRNA
is called **transcription**

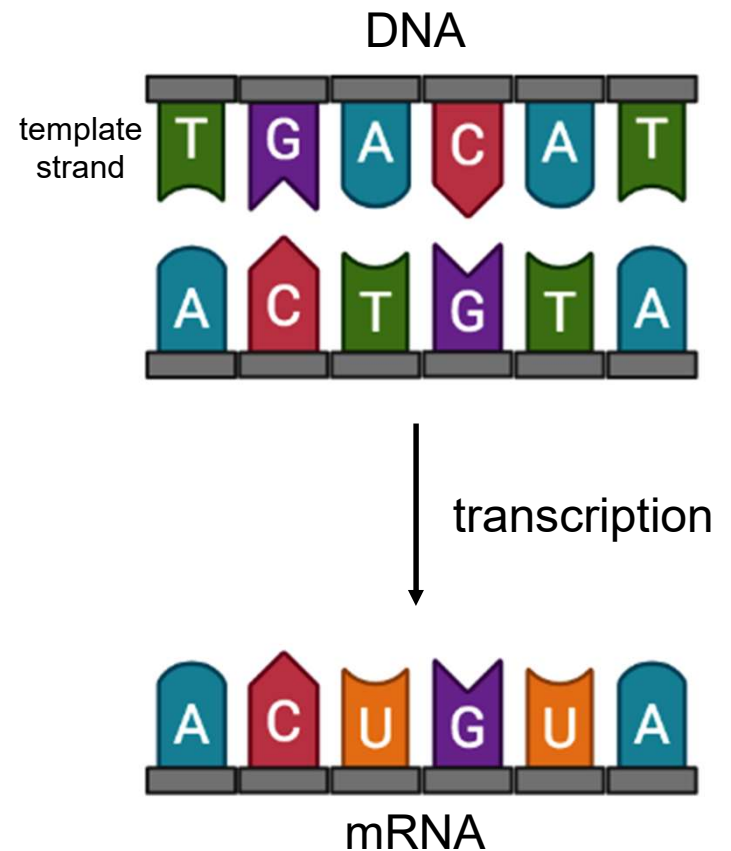
Step 2: RNA → protein
is called **translation**

- DNA codes for proteins/traits: STEP 1 = transcription

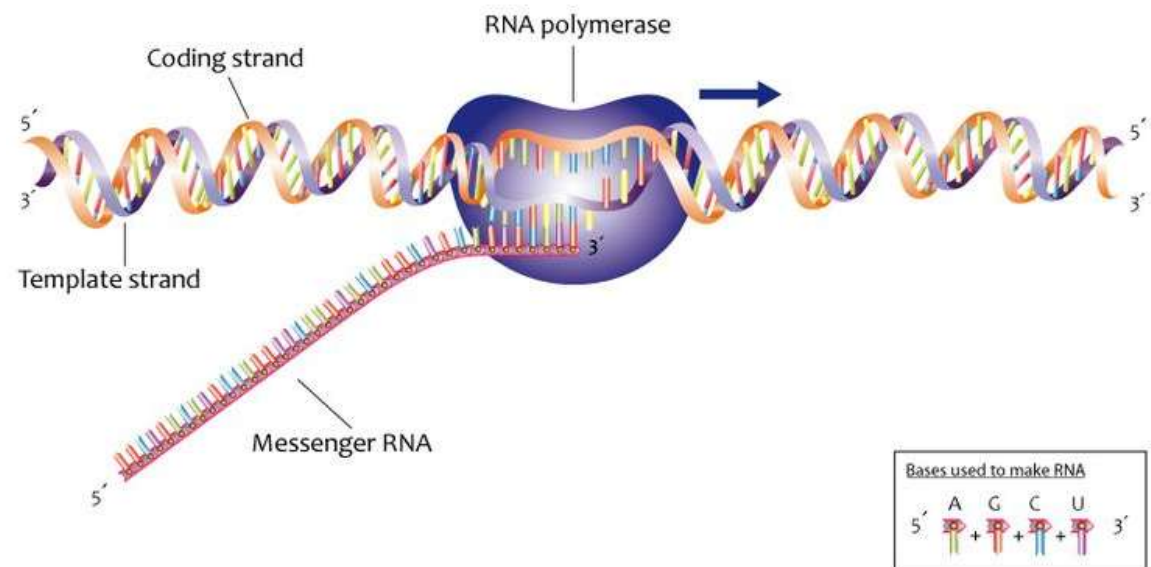
– **Transcription = DNA → mRNA**

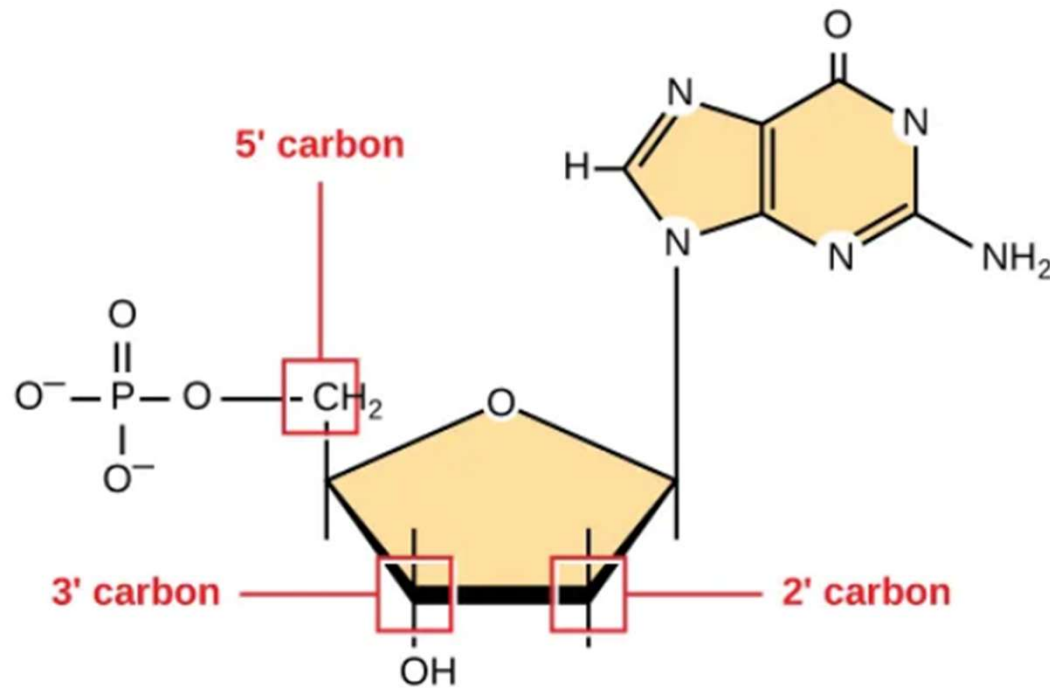
- the information in DNA is copied to RNA inside the nucleus
- the RNA is acting as the middle man, so it is called **messenger RNA (mRNA)**

– *there are other types of RNA we'll bring up soon*



- How do we actually transfer info from DNA to RNA?
- Transcription has 3 stages
 - **1. Initiation:** enzymes unzip the DNA double helix starting at the **promoter** (signals the start of a gene), exposing the **template strand** (the strand that will be copied)





A nucleotide has:
a nitrogenous base, a
pentose sugar, and one or
more phosphate groups.

Carbon residues in the pentose are numbered 1' through 5' (the prime distinguishes these residues from those in the base, which are numbered without using a prime notation). The base is attached to the ribose's 1' position, and the phosphate is attached to the 5' position. When a polynucleotide forms, the incoming nucleotide's 5' phosphate attaches to the 3' hydroxyl group at the end of the growing chain. Two types of pentose are in nucleotides, deoxyribose (found in DNA) and ribose (found in RNA).

- Transcription has 3 stages
 - **2. Elongation:** the enzyme **RNA polymerase** moves along the template strand pairing RNA with DNA (e.g. A with U & C with G), lengthening the mRNA strand

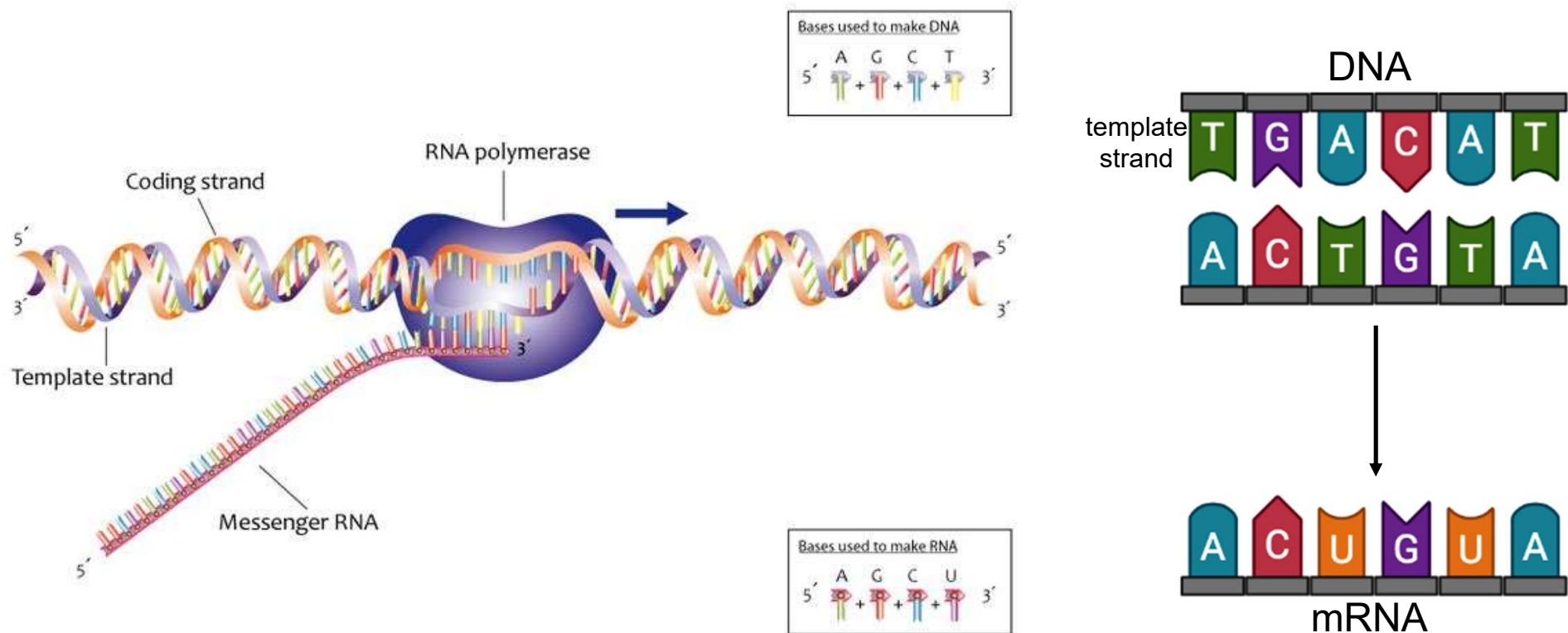
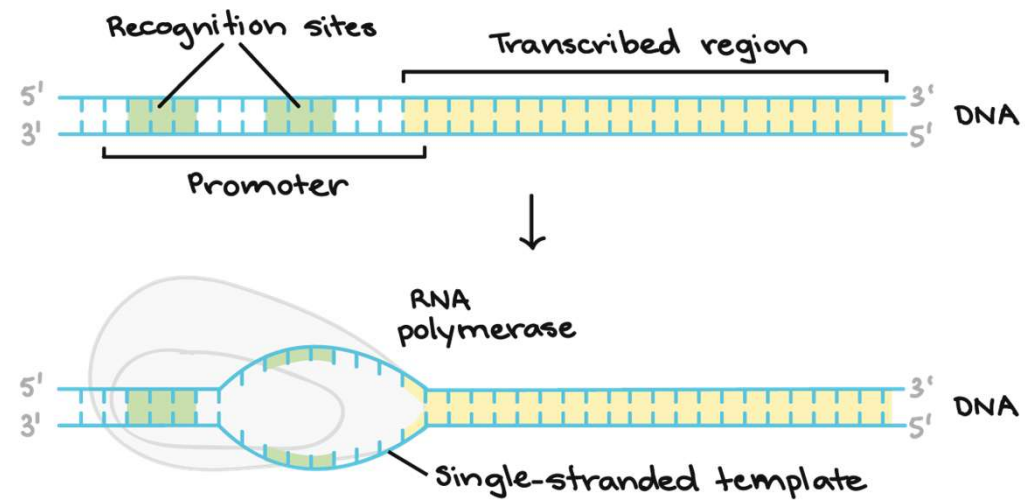
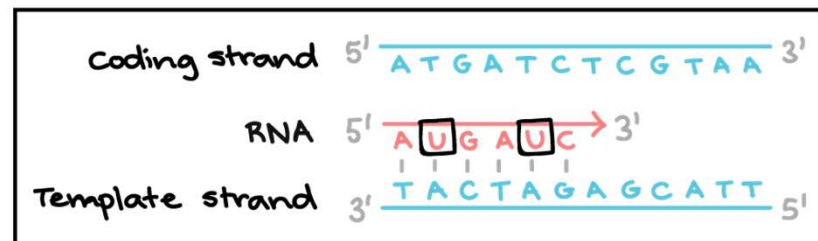
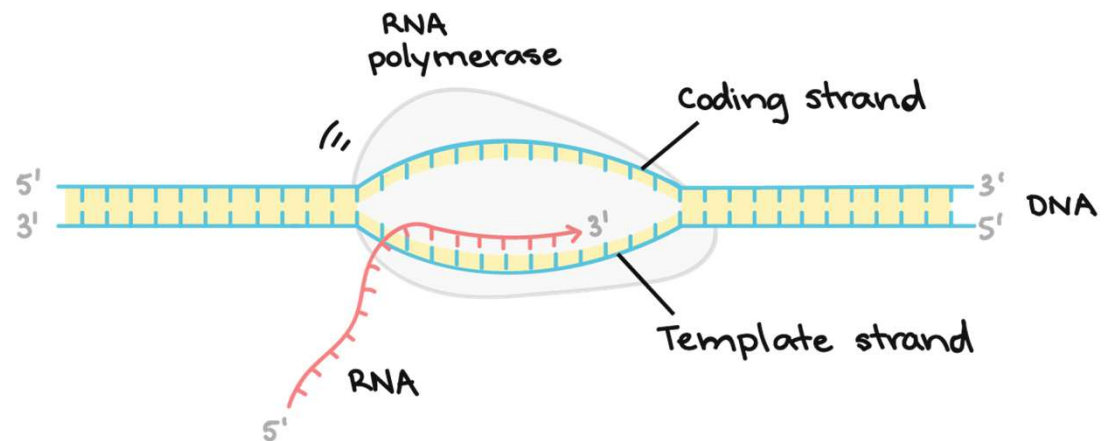


Figure: [https://commons.wikimedia.org/wiki/File:Process_of_transcription_\(13080846733\).jpg](https://commons.wikimedia.org/wiki/File:Process_of_transcription_(13080846733).jpg)

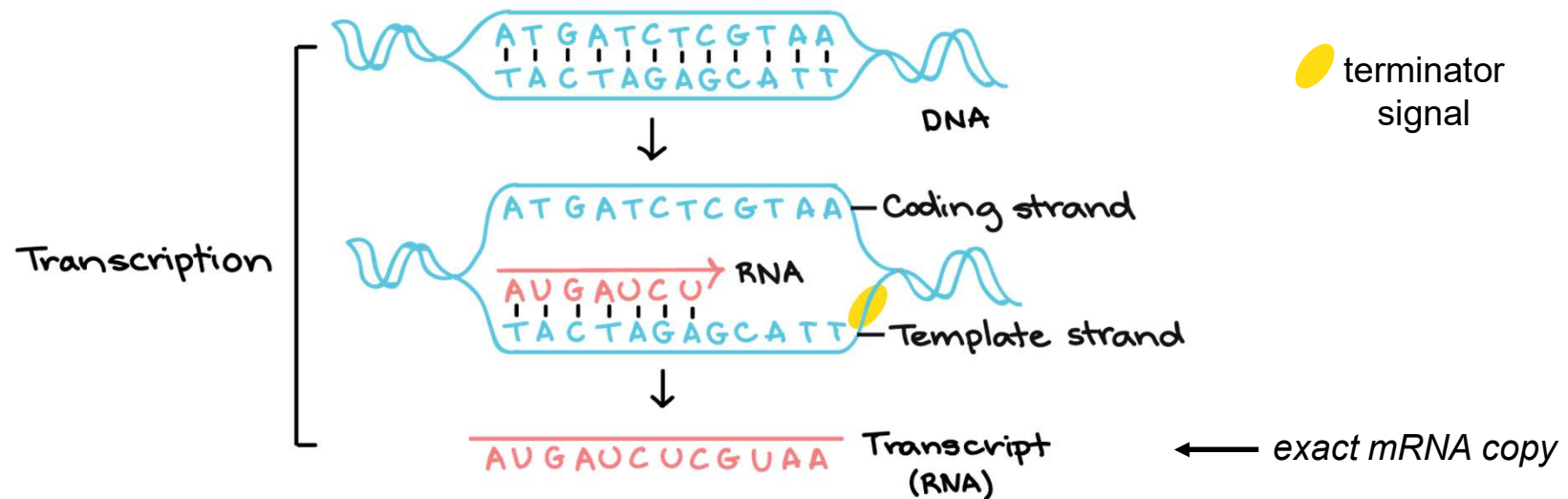
transcription
initiation



transcription
elongation



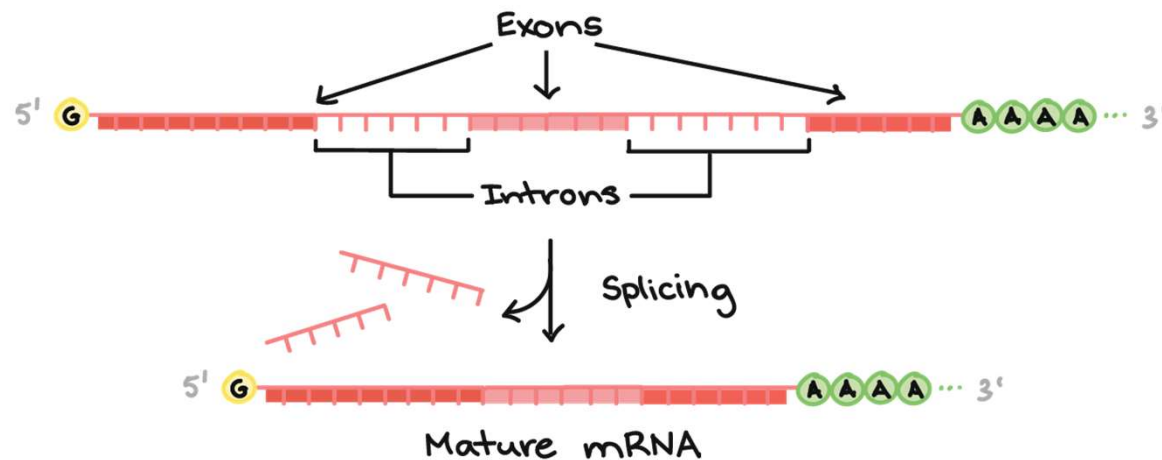
- Transcription has 3 stages
 - **3. Termination:** when the **terminator signal** is reached RNA, DNA, & RNA polymerase fall apart
 - DNA zips back up to a double strand
 - we now have an exact mRNA copy of a gene



Transcription



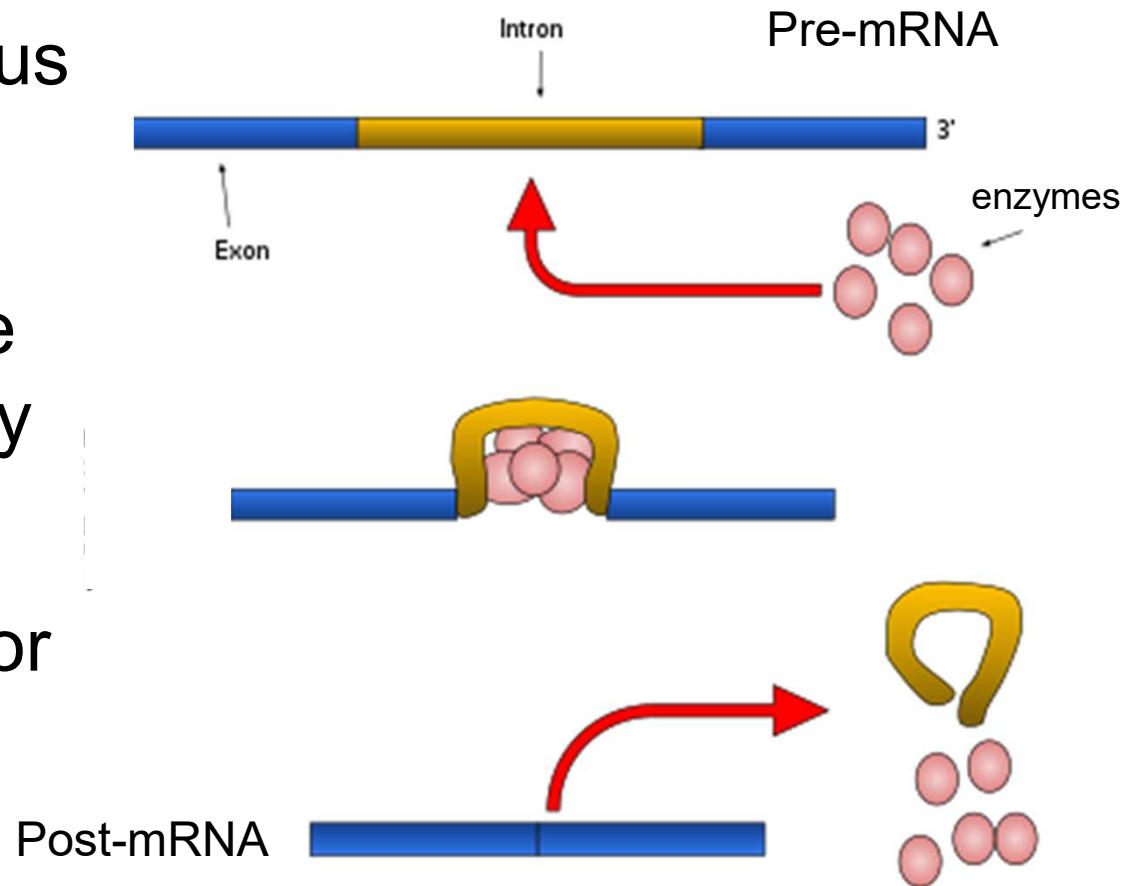
- We now have an exact copy of the gene, but mRNA is not ready to leave the nucleus yet
 - A gene has 2 kinds of segments: introns & exons
 - **exons** = truly coding nucleotides in DNA or RNA
 - **introns** = nucleotides that are not part of the code – must be cut out of mRNA after it's made



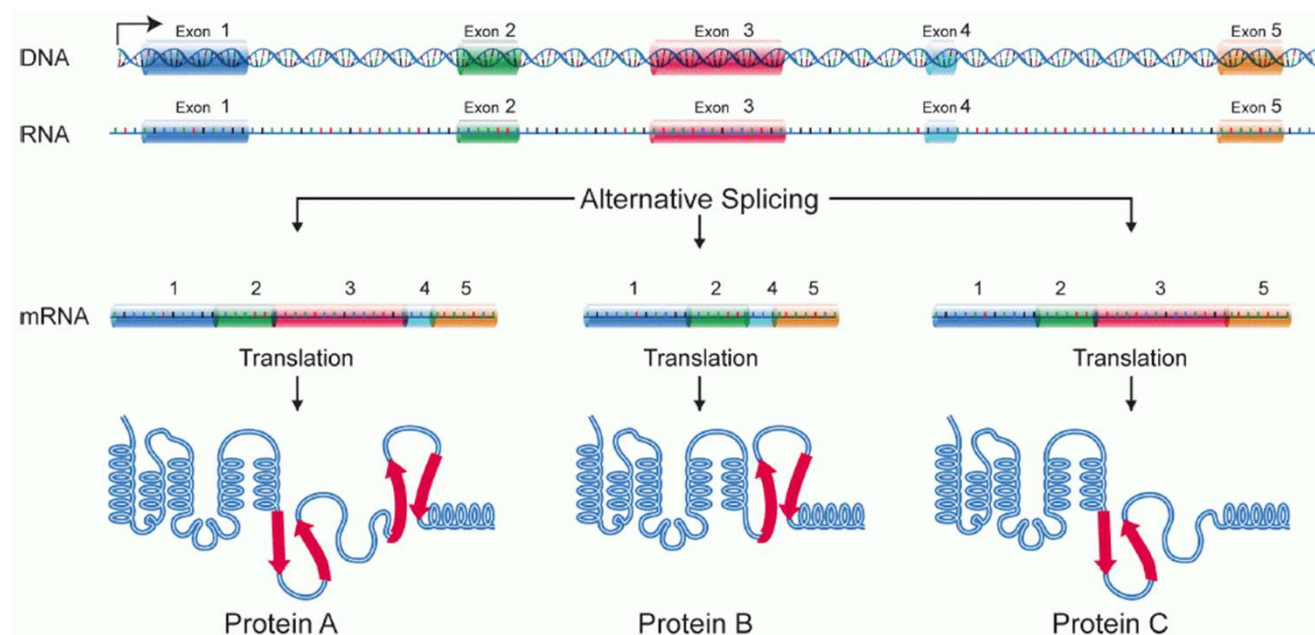
The process of cutting out introns = **RNA splicing**

■ RNA Splicing

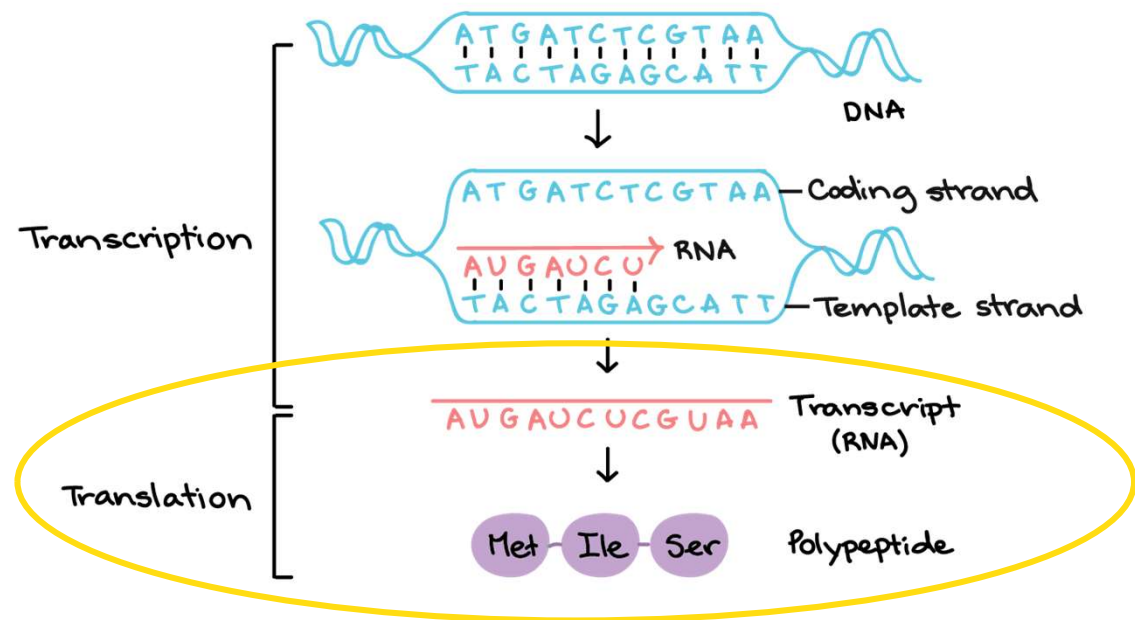
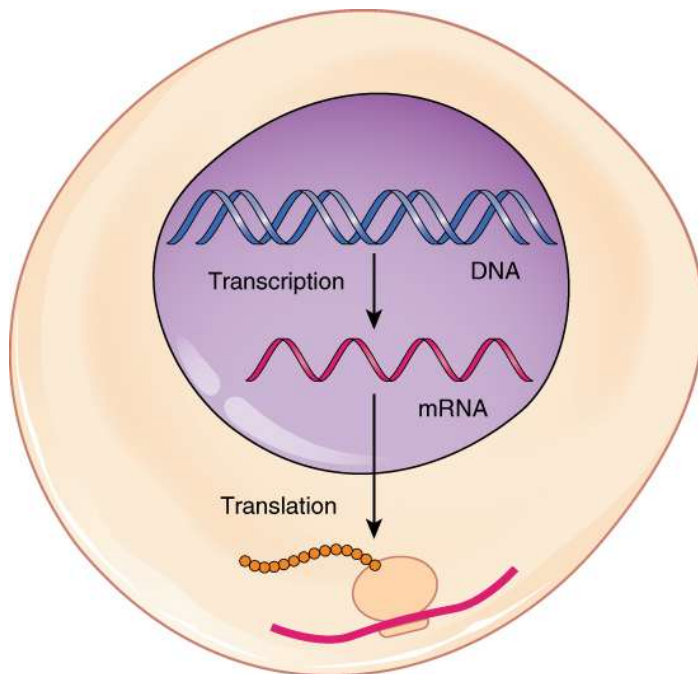
- Immediately after transcription but before splicing, we have **pre-mRNA** – not spliced yet
- Enzymes in nucleus cut out the introns & splice together the exons to make a full strand of only coding mRNA
 - called finished or **post-mRNA**



- Seems inefficient to carry around DNA we don't need – why would RNA splicing be useful?
 - Exons aren't always spliced back together exactly
 - This means the RNA splicing process can allow for 1 gene to make multiple different versions of a protein – very useful!



- DNA codes for proteins/traits: STEP 2 = translation
 - **Translation = mRNA → protein**
 - mRNA moves out of the nucleus into the cytoplasm to direct protein synthesis (make a protein/polypeptide)



- *Remember: **proteins** (polymers) are made of **amino acids** (monomers)*
 - *Remember: there are 20 different amino acids*
 - different amino acid sequence = different protein

Essential	Nonessential
Histidine	Alanine
Isoleucine	Arginine
Leucine	Asparagine
Lysine	Aspartate
Methionine	Cysteine
Phenylalanine	Glutamate
Threonine	Glutamine
Tryptophan	Glycine
Valine	Proline
	Serine
	Tyrosine

Amino acids with different R side chain

AMINO ACID		
Nonpolar, aliphatic R groups		Glycine
		Alanine
		Valine
		Leucine
		Methionine
		Isoleucine
Polar, uncharged R groups		Proline
		Serine
		Cysteine
		Threonine
		Asparagine
		Glutamine

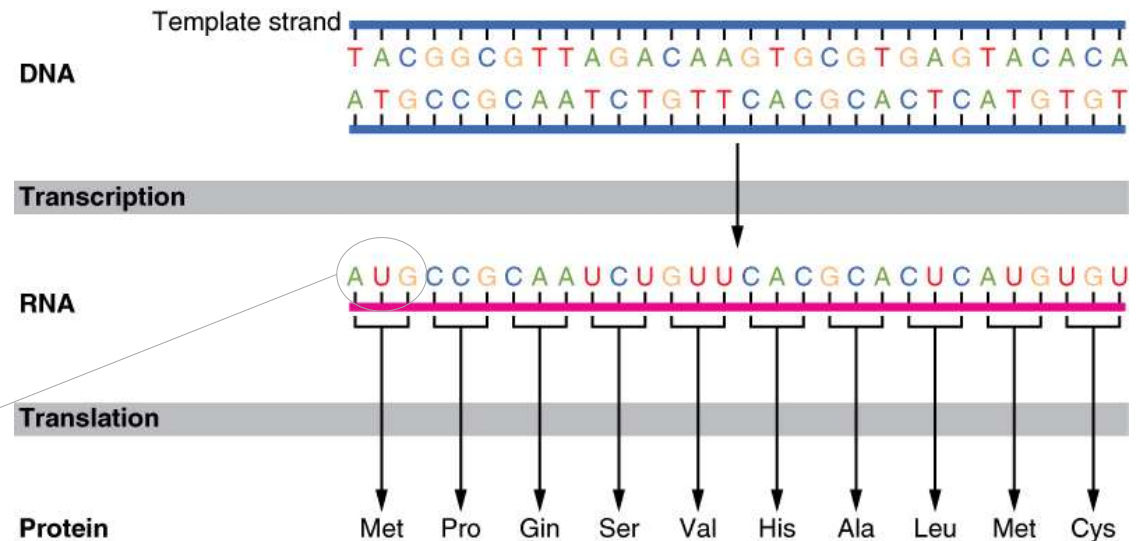
AMINO ACID		
Positively charged R groups		Lysine
		Arginine
		Histidine
Negatively charged R groups		Aspartate
		Glutamate
Nonpolar, aromatic R groups		Phenylalanine
		Tyrosine
		Tryptophan

- *Remember:* in transcription, 1 DNA base matches with 1 RNA base
 - *Remember:* there are only 4 possible bases
- During **translation**, we use an mRNA sequence to build an amino acid sequence
 - If there are only 4 RNA bases but 20 amino acids, we can't match up 1 base with 1 amino acid
 - *we'd only use 4 amino acids & have 16 extra*

- We can try to match up every 2 bases of RNA to every 1 amino acid (e.g. AG = 1 amino acid)
 - there are only 16 possible 2-base combinations
 - *we'd only use 16 amino acids & have 4 extra*
- Must use 3 bases to code for 1 amino acid

– e.g. AUG will code for 1 amino acid

- each 3-base code is called a **codon**



- Translation requires use of the genetic code

- Each **codon** specifies that a unique amino acid should be added to build the protein
- A total of **64 codons**

		Second Base					
		U	C	A	G		
First Base	U	UUU } Phe	UCU } Ser	UAU } Tyr	UGU } Cys	U	Third Base
		UUC } Phe	UCC } Ser	UAC } Tyr	UGC } Cys	C	
		UUA } Leu	UCA } Ser	UAA } STOP	UGA — STOP	A	
		UUG } Leu	UCG } Ser	UAG } STOP	UGG — Trp	G	
	C	CUU } Leu	CCU } Pro	CAU } His	CGU } Arg	U	
		CUC } Leu	CCC } Pro	CAC } His	CGC } Arg	C	
		CUA } Leu	CCA } Pro	CAA } Gln	CGA } Arg	A	
		CUG } Leu	CCG } Pro	CAG } Gln	CGG } Arg	G	
	A	AUU } Ile	ACU } Thr	AAU } Asn	AGU } Ser	U	
		AUC } Ile	ACC } Thr	AAC } Asn	AGC } Ser	C	
		AUA } Ile	ACA } Thr	AAA } Lys	AGA } Arg	A	
		AUG — Met or Start	ACG } Thr	AAG } Lys	AGG } Arg	G	
	G	GUU } Val	GCU } Ala	GAU } Asp	GGU } Gly	U	
		GUC } Val	GCC } Ala	GAC } Asp	GGC } Gly	C	
		GUA } Val	GCA } Ala	GAA } Glu	GGA } Gly	A	
		GUG } Val	GCG } Ala	GAG } Glu	GGG } Gly	G	

- Using the genetic code for translation
 - Notice that one codon is a **start codon** – it signals the start of a gene/protein
 - Notice that a few codons are **stop codons** – they signal the end of a gene/protein
 - All other codons specify that an amino acid should be added to the growing protein

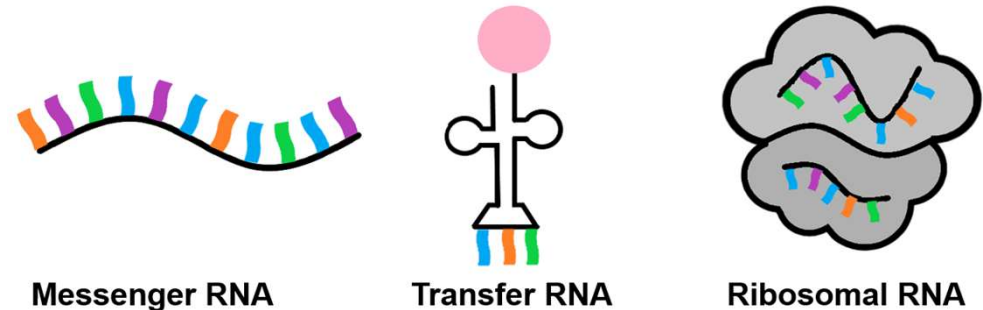
		Second Base					
		U	C	A	G		
First Base	U	UUU } Phe	UCU } Ser	UAU } Tyr	UGU } Cys	U	Third Base
		UUC } Phe	UCC } Ser	UAC } Tyr	UGC } Cys	C	
		UUA } Leu	UCA } Ser	UAA } STOP	UGA } STOP	A	
		UUG } Leu	UCG } Ser	UAG } STOP	UGG } Trp	G	
	C	CUU } Leu	CCU } Pro	CAU } His	CGU } Arg	U	
		CUC } Leu	CCC } Pro	CAC } His	CGC } Arg	C	
		CUA } Leu	CCA } Pro	CAA } Gln	CGA } Arg	A	
		CUG } Leu	CCG } Pro	CAG } Gln	CGG } Arg	G	
	A	AUU } Ile	ACU } Thr	AAU } Asn	AGU } Ser	U	
		AUC } Ile	ACC } Thr	AAC } Asn	AGC } Ser	C	
		AUA } Ile	ACA } Thr	AAA } Lys	AGA } Arg	A	
		AUG } Met or Start	ACG } Thr	AAG } Lys	AGG } Arg	G	
	G	GUU } Val	GCU } Ala	GAU } Asp	GGU } Gly	U	
		GUC } Val	GCC } Ala	GAC } Asp	GGC } Gly	C	
		GUA } Val	GCA } Ala	GAA } Glu	GGA } Gly	A	
		GUG } Val	GCG } Ala	GAG } Glu	GGG } Gly	G	

Using the genetic code for translation

- Notice that there are **64 possible codons** (*combinations of 3, like AAG*)
- But there are only **20 possible amino acids**
- This means that multiple codons code for the same amino acid
 - e.g. UUA & UUG both code for leucine
 - this means the genetic code is **redundant** → **degeneracy**

		Second Base		
		U	C	A
First Base	U	UUU } Phe UUC } UUA } Leu UUG }	UCU } UCC } Ser UCA } UCG }	UAU } UAC } UAA } UAG }
	C	CUU } CUC } Leu CUA } CUG }	CCU } CCC } Pro CCA } CCG }	CAU } CAC } CAA } CAG }
	A	AUU } AUA } Ile AUG }	ACU } ACC } Thr ACA } ACG }	AAU } AAU } AAA }

- How do we physically match RNA to amino acids?

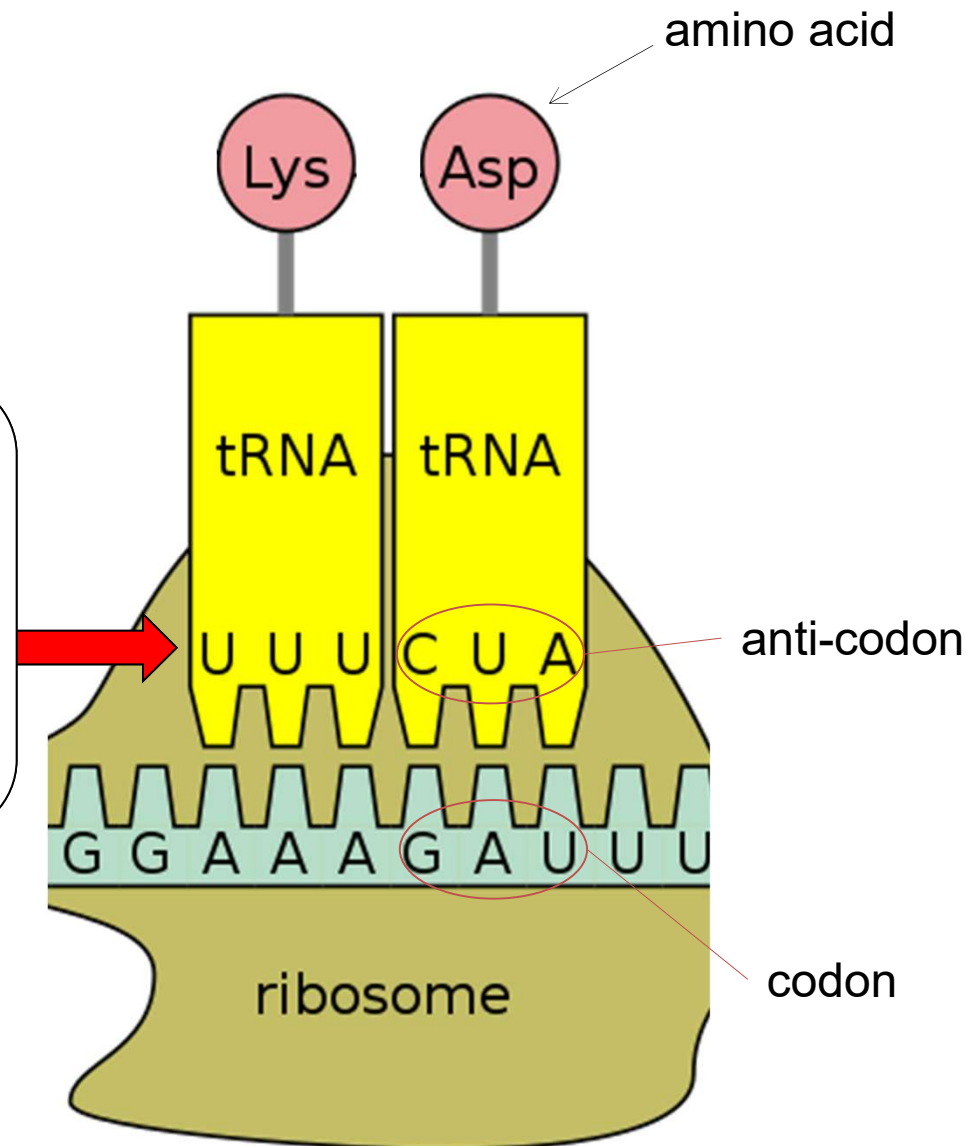


- Key players in translation

- **mRNA** (messenger RNA): brings instructions (in codons) for the amino acid sequence
- **rRNA** (ribosomal RNA): works with proteins to make **ribosomes** (on rough ER) where proteins are made
- **tRNA** (transfer RNA): helps match up each codon to the amino acid it codes for
 - can do this because it has an **anti-codon** (to bind to a codon) & each is attached to a specific amino acid

Transfer RNA (tRNA) molecules translate the mRNA code by linking specific bases on the mRNA with specific amino acids that will be used to build a protein.

The anti-codon is a 3-base sequence that matches up with the 3-base sequence on the mRNA strand. Each 3-base sequence in mRNA—called a codon—matches with a tRNA that carries one particular amino acid.



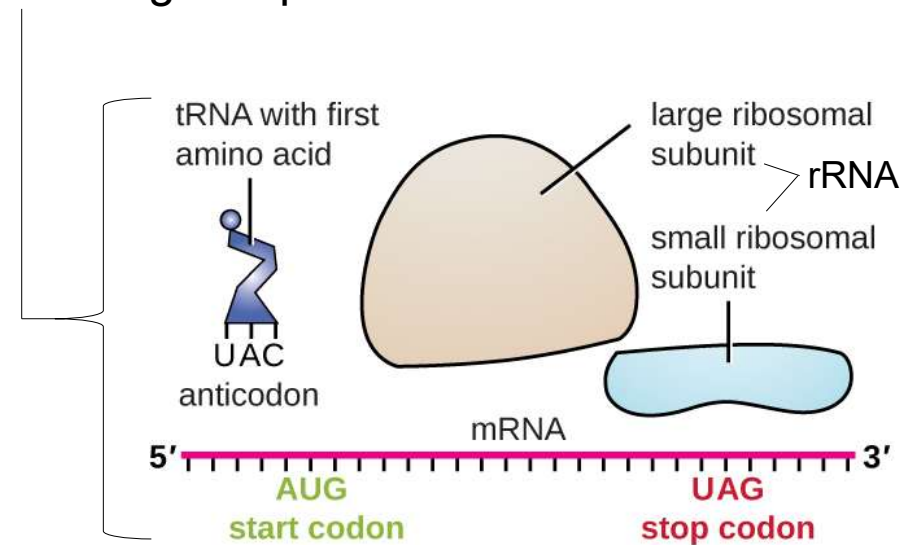
- Translation has 3 stages

- **1. Initiation:**

mRNA, tRNA, & rRNA/ribosomes all come together starting at a **start codon**

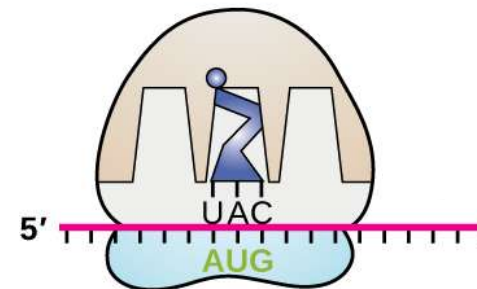
- *the tRNA with the anticodon to the start codon is the first to bind inside the ribosome*

Starting components

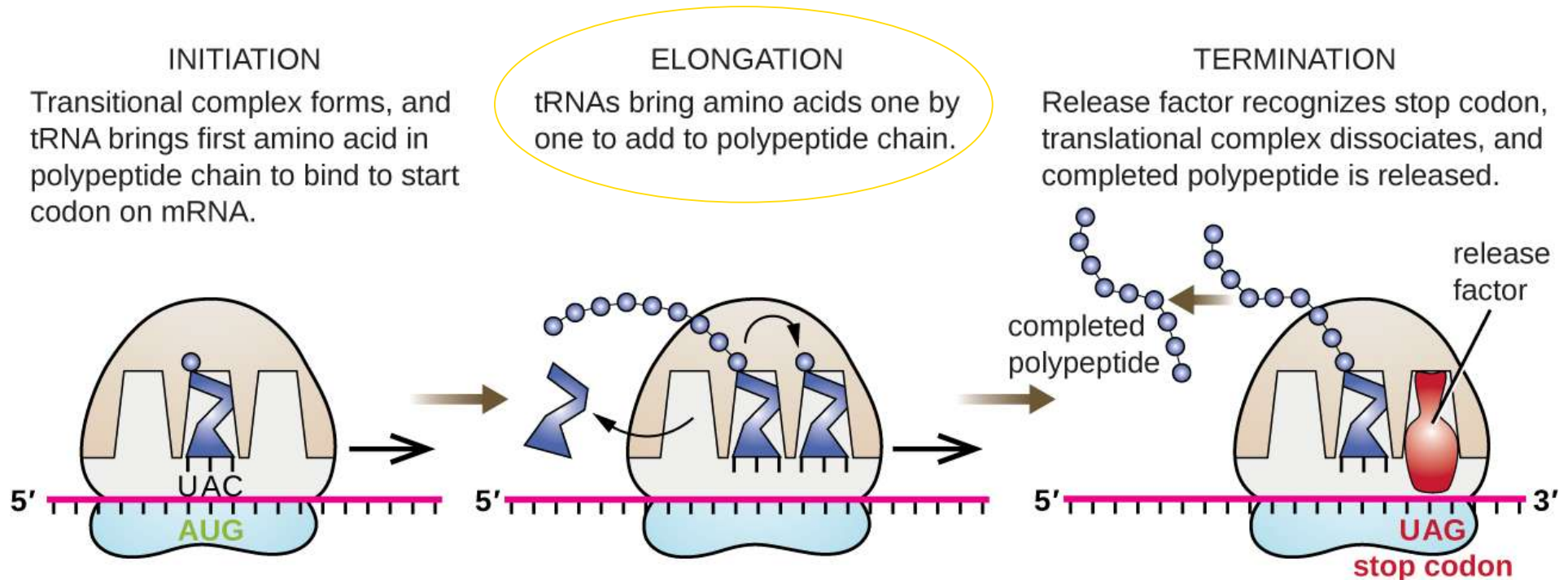


INITIATION

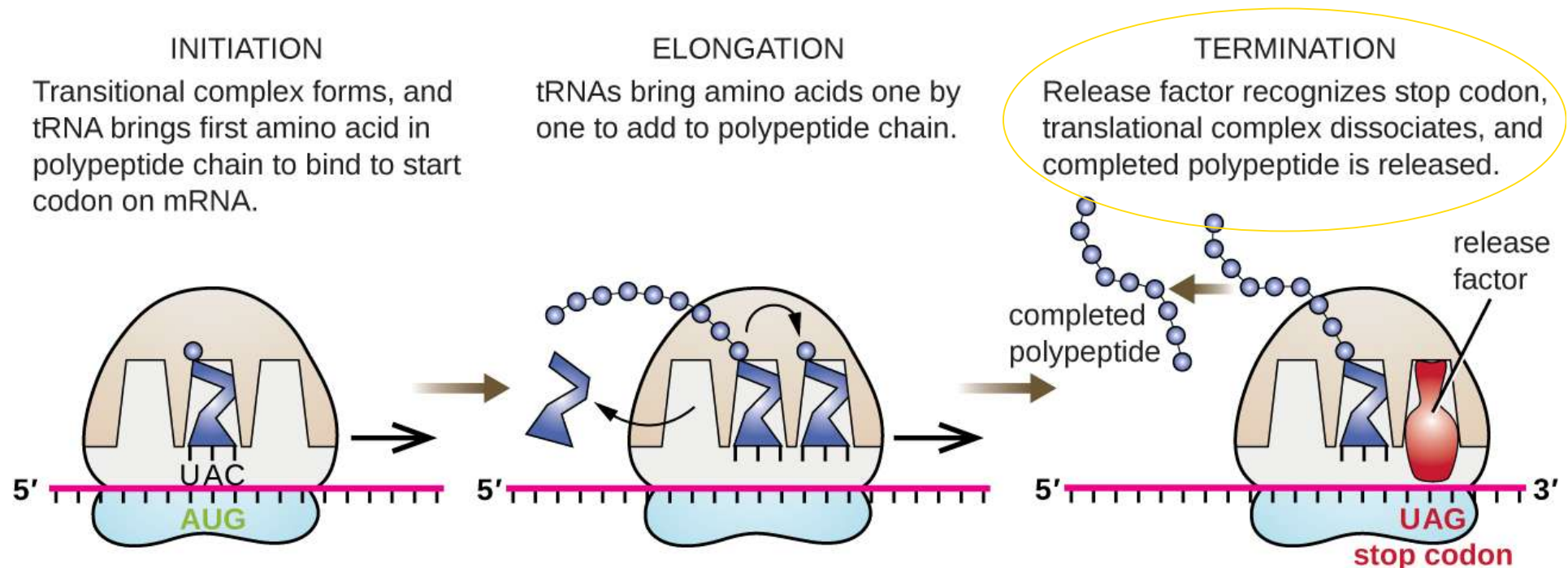
Translational complex forms, and tRNA brings first amino acid in polypeptide chain to bind to start codon on mRNA.



- Translation has 3 stages
 - **2. Elongation:** tRNAs move in & out of the ribosome, leaving their amino acid behind attached to the last amino acid
 - *each time a tRNA leaves, the mRNA shifts over, making space for a new tRNA to bind*



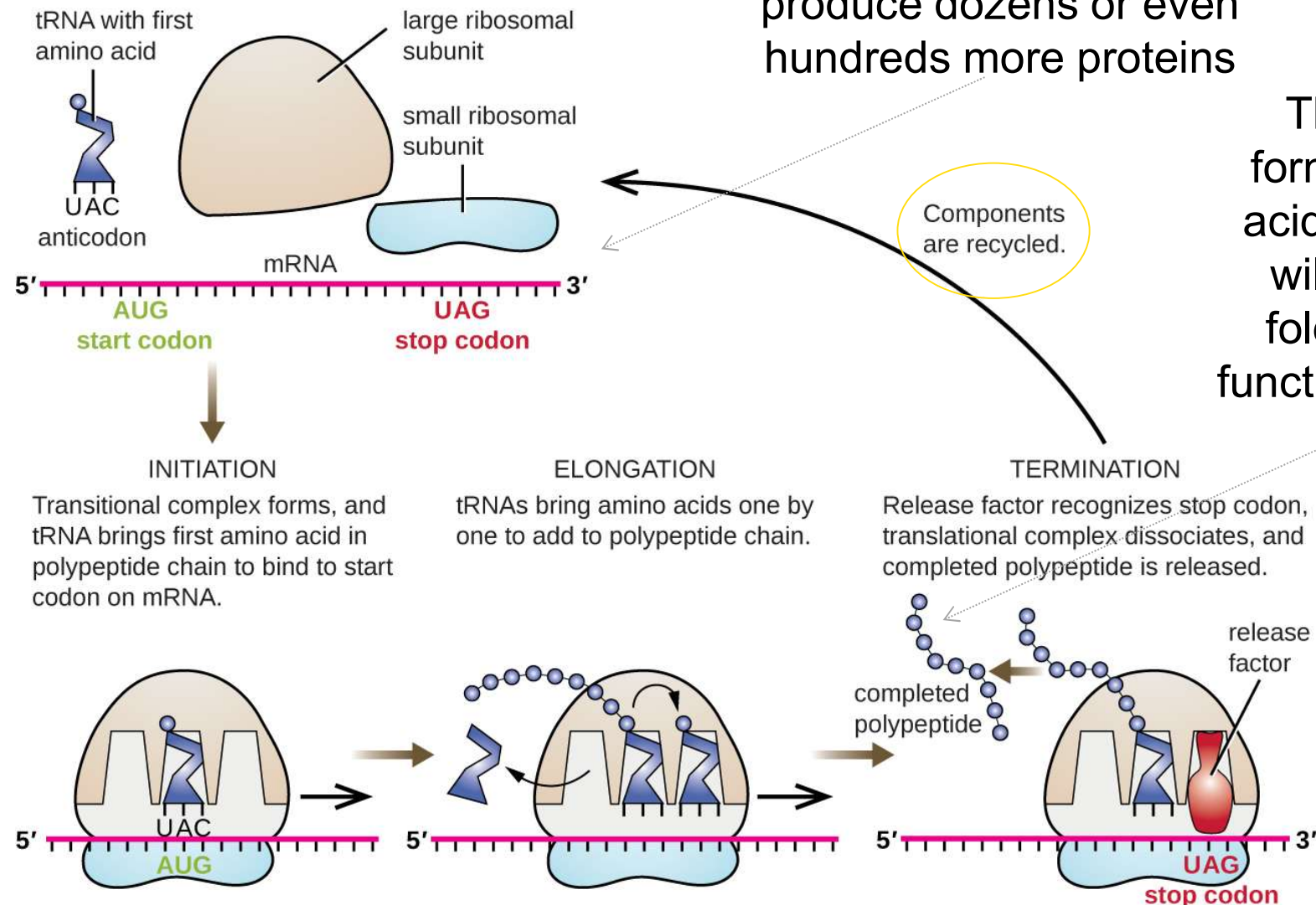
- Translation has 3 stages
 - **3. Termination:** when a **stop codon** is reached, all components separate & the amino acid chain will move through the rough ER & golgi apparatus to be folded & functional



■ After termination:

The mRNA strand can be translated over & over again before it is degraded – can produce dozens or even hundreds more proteins

The newly formed amino acid sequence will become folded into a functional protein



Translation

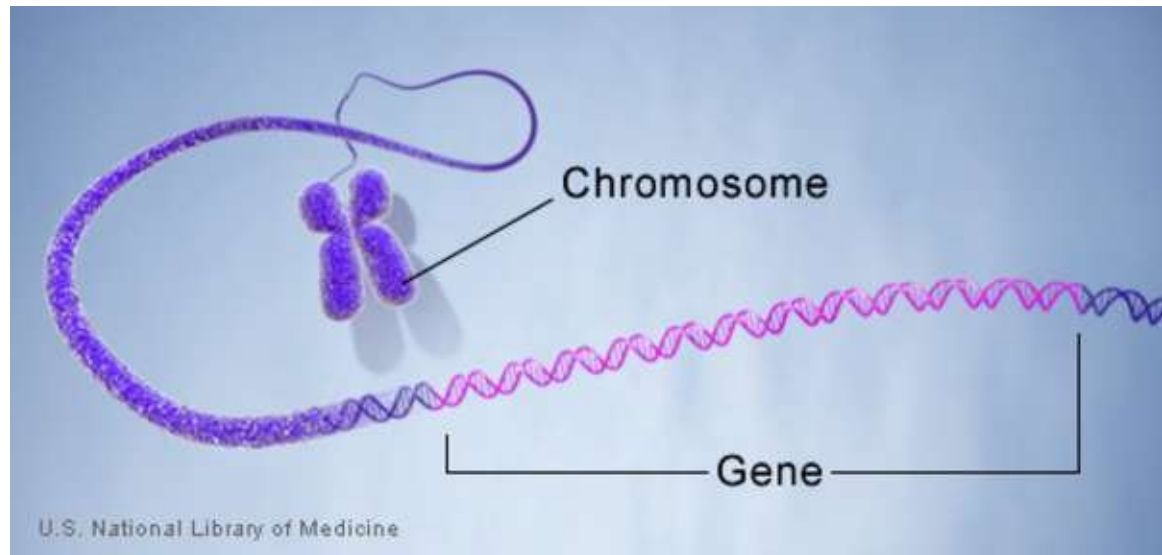




<https://dnalc.cshl.edu/resources/animations/>

How are genes regulated?

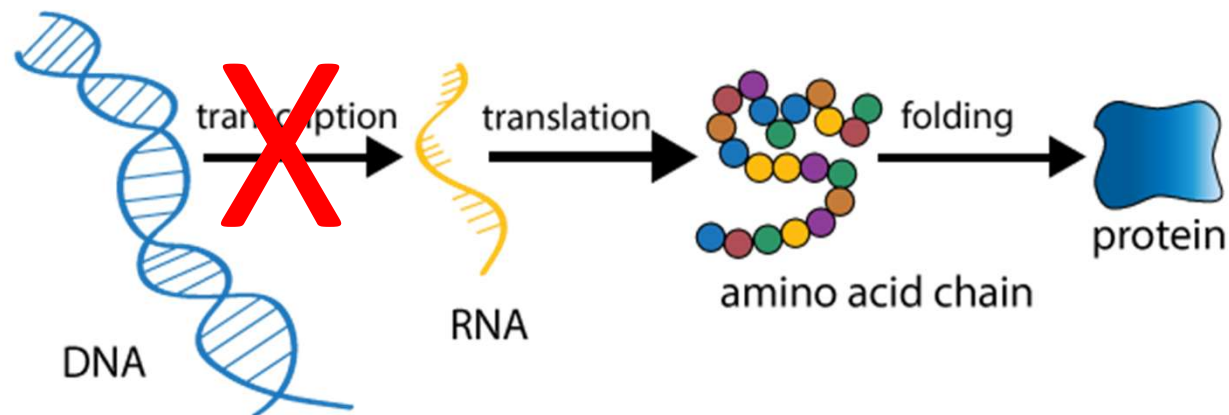
- We now see how our DNA codes for our traits, making us who we are
- BUT: not all of our genes are expressed at all times
 - Humans have around 25,000 pairs of genes (1 from each parent) spread across 23 pairs of chromosomes



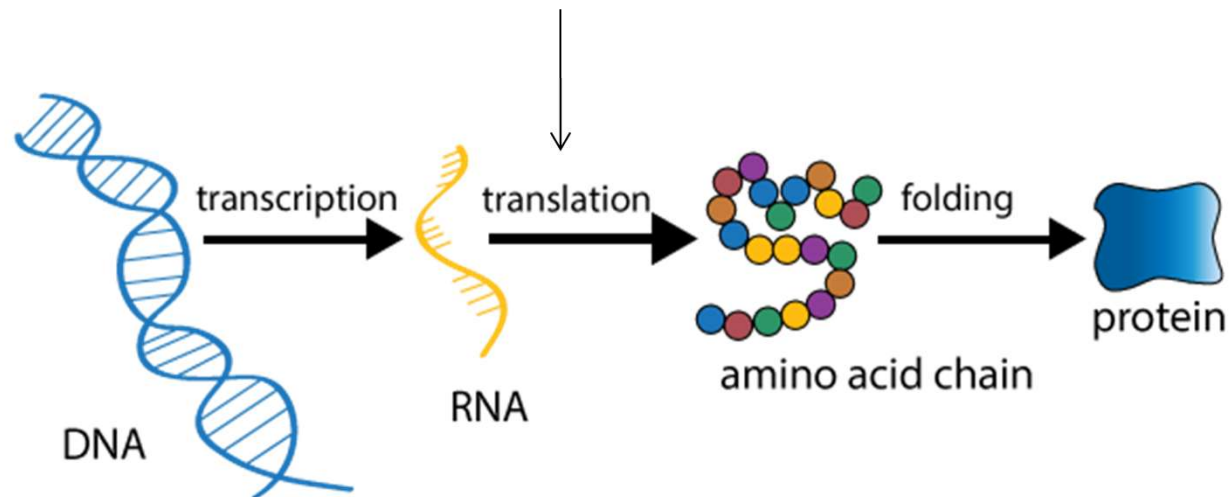
- **Gene expression:**

- Some genes are only expressed at certain times
 - e.g. every human has 2 genes that code for milk proteins during lactation, but only some people produce these proteins at only specific times
- Some are never used by a cell
 - e.g. skin cells never need hemoglobin (for blood)
- Some are always expressed in every cell
 - e.g. cellular respiration / mitochondria enzymes
 - *(all cells need ATP)*

- Cells can regulate gene expression at 3 levels
- **Level 1: Transcription**
 - cells control which genes are actually transcribed from DNA → RNA
 - no transcription = no expression at all (e.g. milk proteins in cells that aren't lactating)

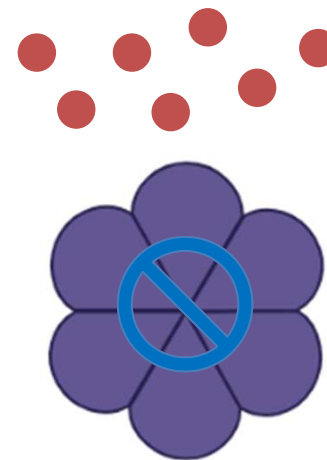
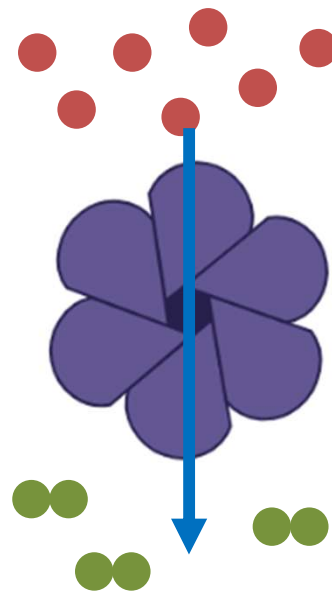


- Cells can regulate gene expression at 3 levels
- **Level 2: Translation**
 - cells control how many proteins are made from mRNA after transcription
 - does 1 mRNA strand make 1 protein & stop, or should it make 100 proteins?



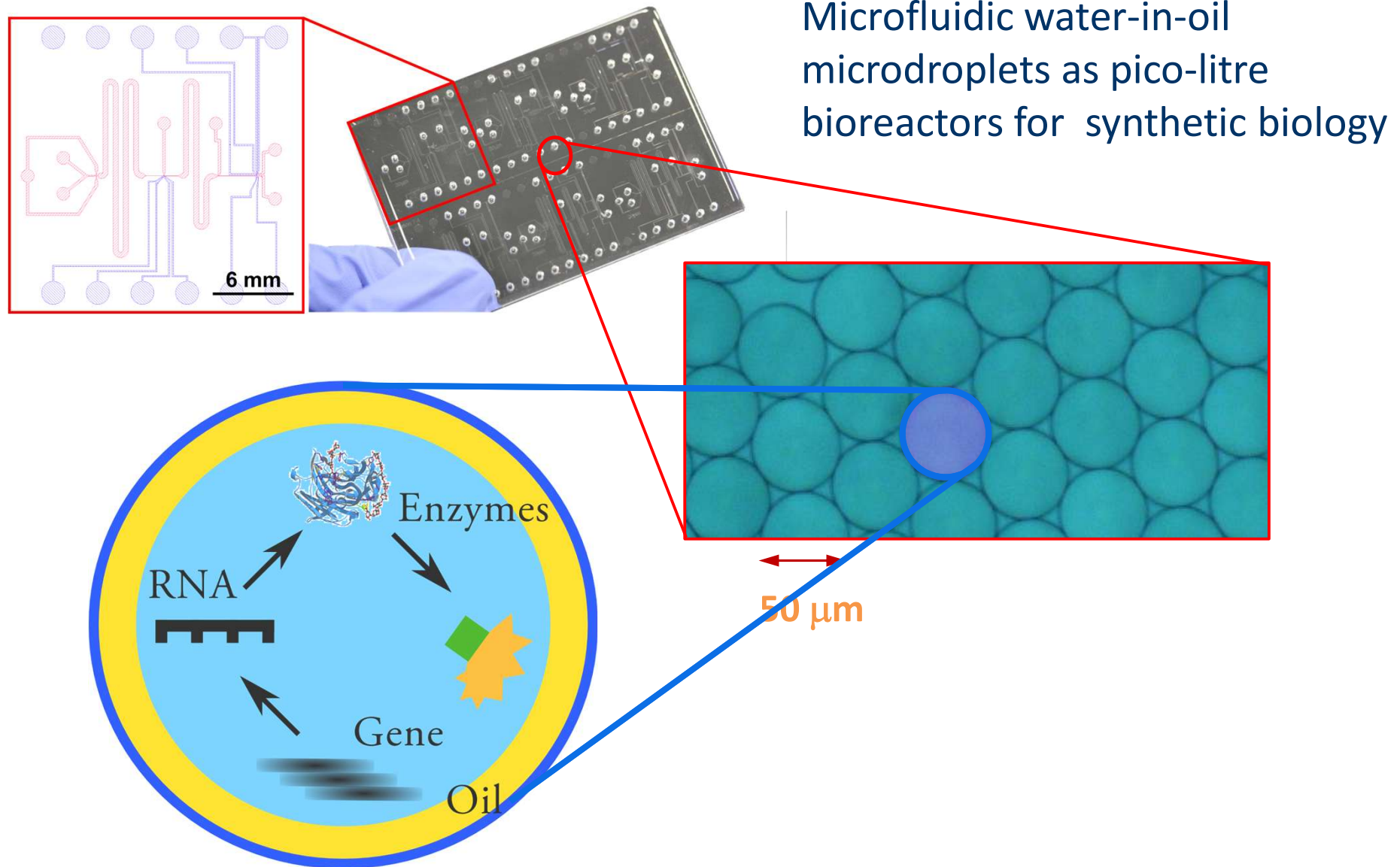
- Cells can regulate gene expression at 3 levels
- **Level 3: Protein activity**
 - *Remember:* cells can control metabolism by activating or inactivating proteins that have already been made

A regularly
functioning
protein



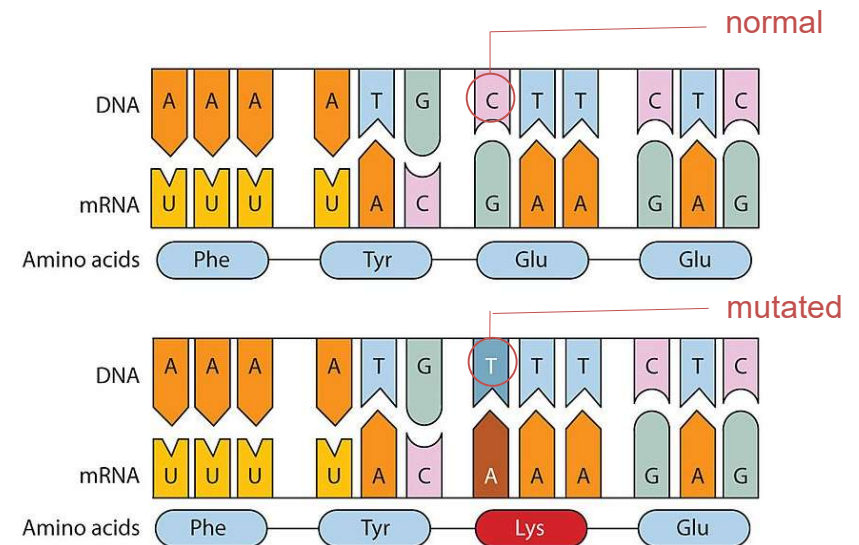
A protein that
has been
inactivated

Biotechnology: Gene expression for enzyme synthesis

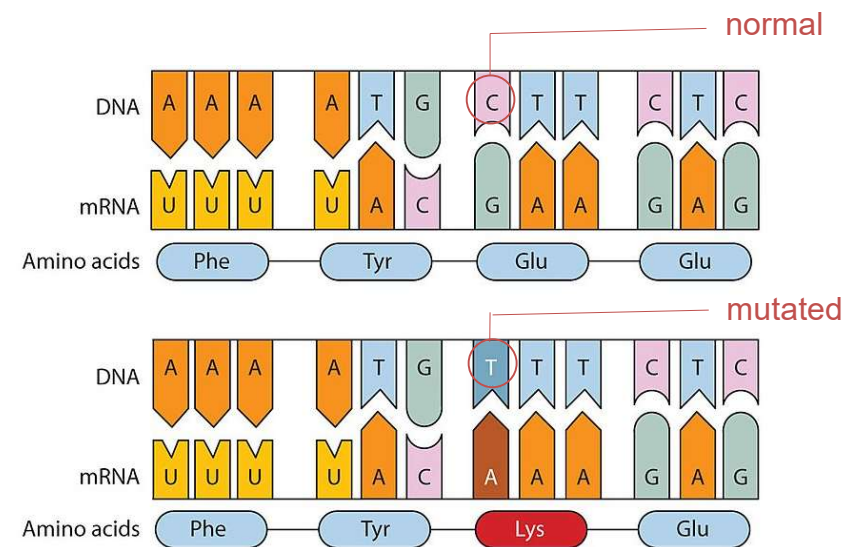


How do mutations affect genes?

- **Mutations** = changes in the sequence of DNA bases
 - Can occur naturally, but are usually caused by carcinogens & mutagens
 - **Mutagen** = a substance that causes mutations
 - **Carcinogen** = a substance capable of causing cancer, usually by mutating DNA (*more on cancer in next chapter*)

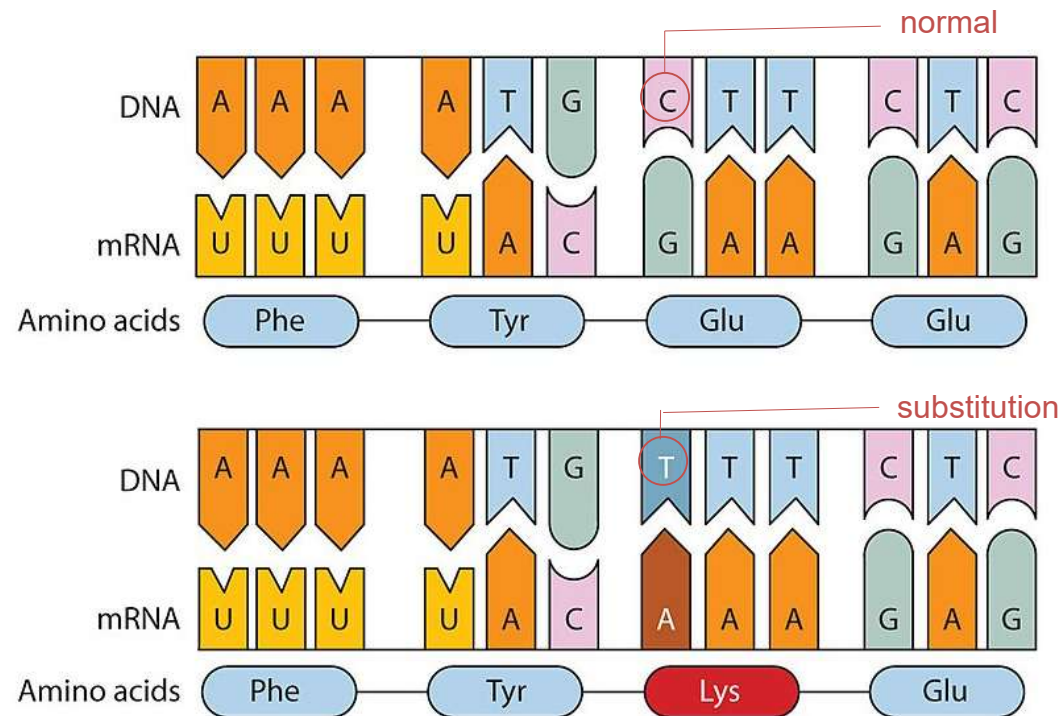


- We come into contact with many carcinogens & mutagens in our environment all the time
- Many types of mutations can occur
 - Sometimes only one base changes
 - e.g. substitution
 - e.g. insertion
 - e.g. deletion
 - Sometimes large pieces of DNA are moved
 - e.g. inversion
 - e.g. translocation



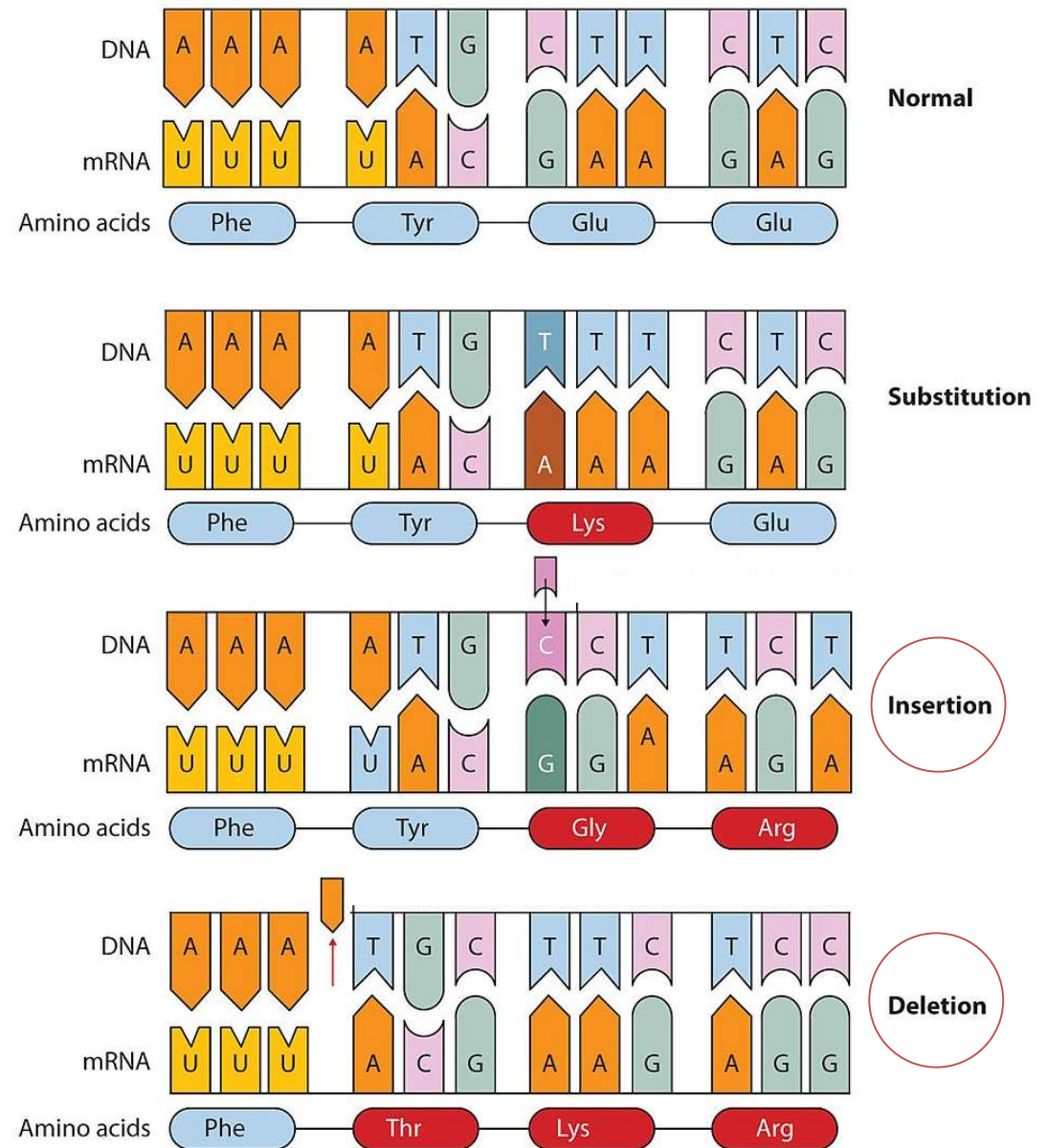
■ Substitution mutations:

- one base is swapped out for a different base
- e.g. an A is swapped (substituted) for a T



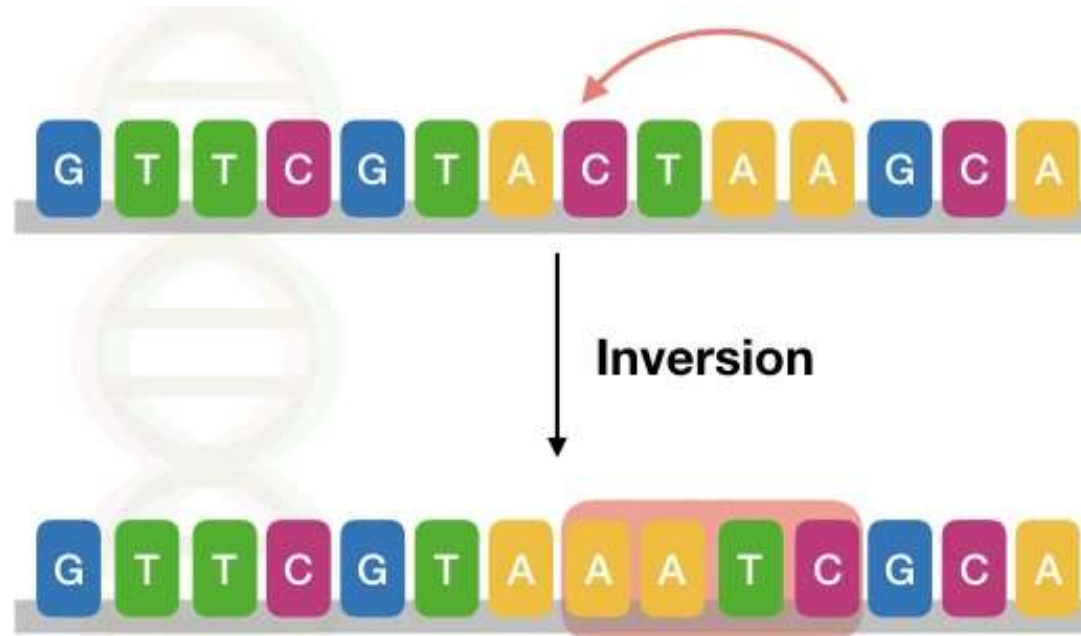
- **Insertion & deletion mutations:**

- 1 or more nucleotides are inserted or deleted into the DNA sequence



- **Inversion mutations:**

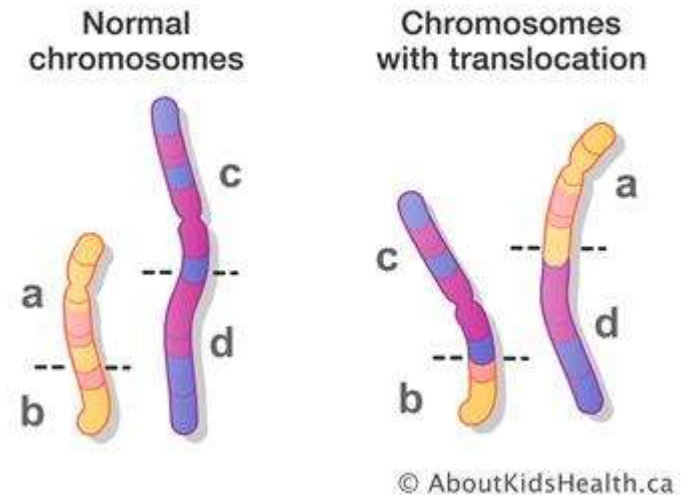
- a piece of DNA is accidentally cut out of a chromosome, gets turned around, & is re-inserted into the gap upside down



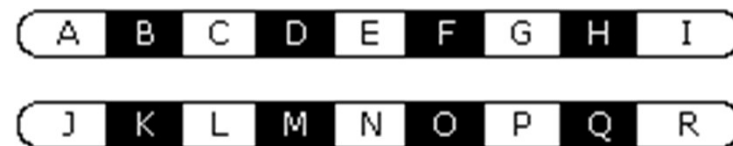
■ Translocation mutations:

- a piece of DNA (often very large) is accidentally removed from 1 chromosome & attached to another chromosome

- only time more than one chromosome is involved



2 different healthy chromosomes



2 different chromosomes after translocation



A silly summary of the different types of mutations:

Normal



BEAST

Substitution



FEAST

Insertion



BREAST



Deletion



BEST



A

Inversion



BEATS

- What are the effects of mutations?
 - **Inversion & translocation** mutations involve large pieces of DNA
 - These may be neutral if the entire gene is moved from one place to another – very rare!
 - Most common: a gene is split in 2
 - when transcription occurs, the message is cut short & no functional protein can be made: harmful

Normal vs. Mutation		Hypothetical visualization of effects	
Normal DNA codons		THE ONE BIG FLY HAD ONE RED EYE	
Translocation		THE ONE BIG F	

- What are the effects of mutations?
 - One **insertion** or **deletion** mutation causes all following nucleotides to shift over – this changes every codon after the mutation = **frameshift**
 - changing the codons means changing the amino acids coded for during translation, & almost always = no functional protein: harmful

Normal vs. Mutation	Hypothetical visualization of effects
Normal DNA codons	THE ONE BIG FLY HAD ONE RED EYE
Insertion (of Q)	THE ONE BQI GFL YHA DON ERE DEY E..
Deletion (of G)	THE ONE BIF LYH ADO NER EDE YE.

■ What are the effects of mutations?

– **Substitution**

- *Remember:* the genetic code is **redundant**
 - e.g. UUA & UUG both code for leucine
- the codon is changed but may still code for the same amino acid: the mutation can be neutral

Normal vs. Mutation		Hypothetical visualization of effects	
Normal DNA codons		THE ONE BIG FLY HAD ONE RED EYE	
Substitution (S for G)		THE ONE BIS FLY HAD ONE RED EYE	

If BIG & BIS both code for the same amino acid, the final protein is unchanged & perfectly functional

■ What are the effects of mutations?

– **Substitution**

- If the mutation causes a different amino acid to be coded for, the mutation can lead to a nonfunctional protein: harmful
- It may even code for a premature stop codon, which leads to an incomplete protein: harmful

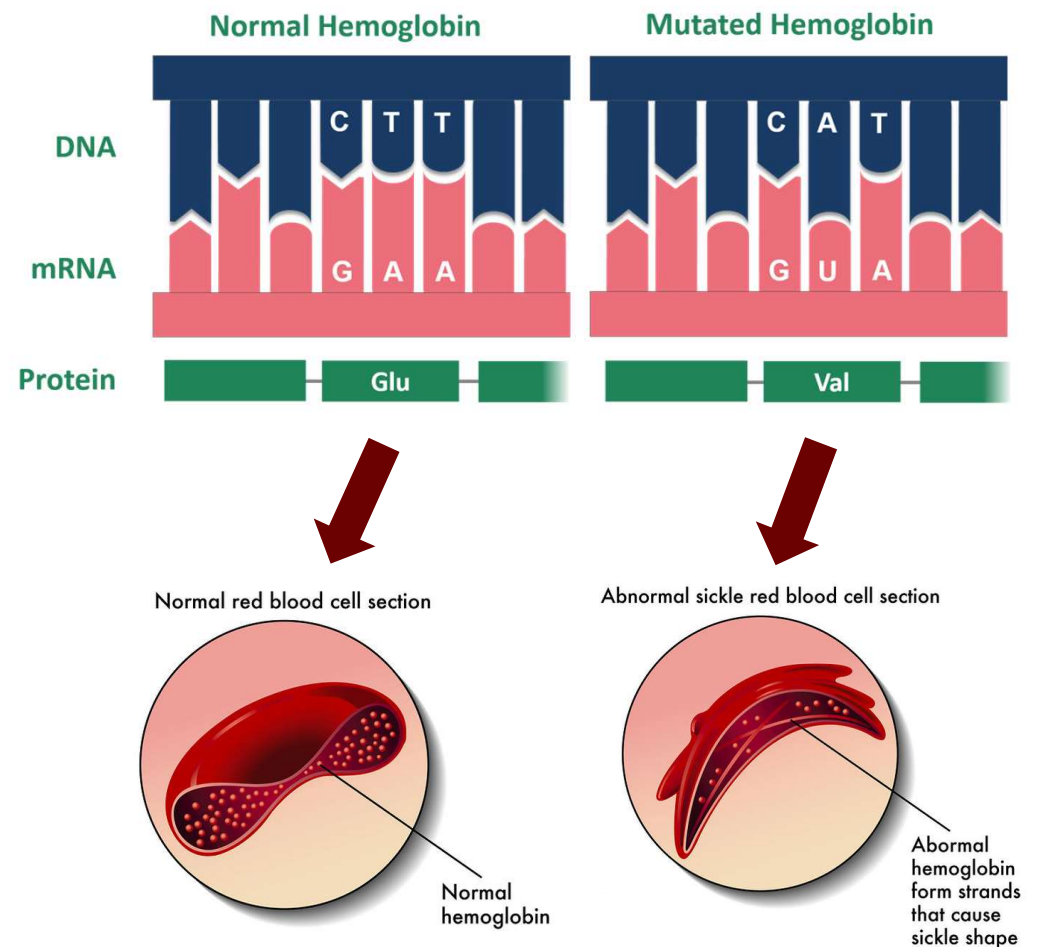
Normal vs. Mutation	Hypothetical visualization of effects
Normal DNA codons	THE ONE BIG FLY HAD ONE RED EYE
Substitution (S for G)	THE ONE BIS FLY HAD ONE RED EYE

If BIG & BIS code for different amino acids, the final protein is changed & may not be functional

- Some mutations can be beneficial: e.g. 1 substitution on the hemoglobin gene codes for proteins that cause **sickle cell anemia**

- Can be harmful
- However: sickle cell anemia provides resistance to malaria

- for people that live in malaria-prone areas of the world, this mutation is beneficial



Biotechnology: enzyme evolution by mutating genes

Microfluidic water-in-oil microdroplets as pico-litre bioreactors for synthetic biology

